North Eastern Mindanao State University
Republic of the Philippines

A DECISION SUPPORT SYSTEM ARCHIVING HUB: TRANSFORMING THESIS AND FEASIBILITY STUDY USING LLAMA

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Dedication

With deep gratitude and love, we, the researchers, wholeheartedly dedicate this thesis to our beloved families, whose unwavering support, sacrifices, and encouragement have been the cornerstone of our strength and determination.

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Abstract

This study focuses on the development and evaluation of the Decision Support System (DSS) Archiving Hub, a web-based platform designed to improve the storage, retrieval, and organization of theses and feasibility studies at North Eastern Mindanao State University (NEMSU) – Tandag Campus. The platform addresses inefficiencies in traditional manual archiving by providing a centralized digital repository that supports structured access to academic outputs. A key feature of the system is the integration of LLaMA, a language model utilized exclusively to generate research gap suggestions from thesis abstracts, helping students avoid topic duplication and guiding future research directions.Developed using modern web technologies and grounded in usercentered design, the DSS Archiving Hub introduces role-based access, smart filtering, and AI-assisted research guidance. The study adopts a descriptivedevelopmental research design, involving 363 respondents—344 students and 19 instructors—to evaluate system performance based on ISO 25010 software quality metrics.Results indicate high user satisfaction across usability, functional suitability, reliability, performance efficiency, perceived ease of use, and security. Findings highlight the DSS Archiving Hub as a transformative academic tool that improves institutional research management, enhances research integrity, and supports data-driven educational planning. The study not only addresses the limitations of manual archiving systems but also validates a scalable and intelligent digital solution tailored for academic environments.

Keywords: Decision Support System, Thesis Archiving, Feasibility Study Repository, Research Gap Detection, LLaMA, ISO 25010, Academic Digital Platform, NEMSU.

CHAPTER I

INTRODUCTION

Background of the Study

The academic works found in university institutions at NEMSU Tandag Campus, such as thesis and feasibility studies, were vital intellectual resources. However, University institutions still relied manual methods for archiving and retrieving these documents, which led to inefficiencies, limited accessibility, and loss in documentation. This outdated process made it challenging for students and faculty members to access valuable research materials easily for reference and further study. This study proposed the creation of a Decision Support System Archiving Hub using Llama to transform how academic works like theses and feasibility studies were archived and accessed. The system aimed to streamline document submission, storage, and retrieval processes by integrating modern digital archiving technologies to ensure fast and secure handling of academic resources.

The Decision Support System Archiving Hub powered by Llama provided a centralized platform for systematically categorizing academic papers and making them readily available to users. Furthermore, the system prioritized data privacy by implementing Access to the robust measures to protect sensitive academic information and intellectual property. archived works was carefully regulated, ensuring that only authorized users can retrieve documents. User access was managed through account-based authentication

and structured system roles to ensure that only authorized individuals could upload, view, or manage academic resources, thereby minimizing the risk of accidental loss or misuse. Additionally, the system also organized uploaded academic work in a structured manner, avoided topic duplication by filtering titles and identified potential gaps for future research. These capabilities enabled users to locate relevant resources and guided institutions in managing research trends with the integration of Llama. This comprehensive approach not only improved accessibility but also supported better academic planning and resource management. The system's goal was to provide a comprehensive, efficient, and secure solution for academic document management, thereby enhancing the accessibility, organization, and protection of institutional research data.

Other studies express concerns regarding the transparency of the manual method of archiving theses and document studies. Researchers and students who seek specific materials in these archives often find traditional navigation methods to be tedious, time-consuming, and frustrating (Nitin Ghadge, 2024). A common issue in university-level research is the loss of data due to poor data management or the departure of essential staff members. Data is frequently stored on personal laptops, flash drives, or other devices that are not part of a centralized system, increasing the risk of mismanagement or loss (Wright et al., 2020). Moreover, confusion surrounding the elements of research papers remains prevalent among students. The key to writing an effective

research paper lies in understanding these common elements and their definitions, though many students still find it challenging (Alkhaqani, 2022).

Another issue is the reliance on traditional teaching materials, which no longer engage students. Despite the availability of online journals and websites, students often prefer unverified articles because they appear quickly in search engine results, neglecting more reliable sources (Febriati et al., 2020). Combining these challenges with a Decision Support System (DSS) Archiving Hub that utilized Llama's advanced capabilities can transformed how academic works such as theses and feasibility studies were archived and accessed, providing an efficient and centralized solution.

Academic institutions have implemented digital solutions for storing documents, but challenges persisted in efficiently managing large volumes of research papers such as theses and feasibility studies. Manual paper-based archiving creates inefficiencies, leading to cumbersome access and retrieval. Previous studies on digital archiving systems often overlooked the need for intelligent data organization and decision support tools required to enhance accessibility to academic research. To address these shortcomings, this study presented the Decision Support System Archiving Hub enhanced by Llama, a cutting-edge platform aimed at revolutionizing the management and retrieval of academic works. By leveraging Llama's advanced capabilities, the system will assist in identifying gaps in existing academic work, helping users locate relevant resources while guiding institutions in managing research trends. This

platform offered a user-friendly experience, allowing students and faculty to access research materials efficiently through a web-based interface. The Decision Support System Archiving Hub ensured rapid retrieval, secure storage, and organized archiving, ultimately transforming the accessibility and usability of academic archives.

In addressing the inefficiencies of traditional academic archiving systems at NEMSU Tandag Campus, this study proposed the Decision Support System Archiving Hub, a decision support system that redefined how theses and feasibility studies were stored and accessed. The system introduced features like categorization, and data-driven insights powered by Llama's tools to identify gaps in existing research. Unlike conventional systems, the Decision Support System Archiving Hub integrated decision support tools to provide analytics on academic trends, enhancing both resource management and institutional planning. This innovative solution was tailored to the unique needs of academic research at NEMSU Tandag Campus, transforming storage and accessibility while fostering a more organized and efficient research environment.

Theoretical Framework

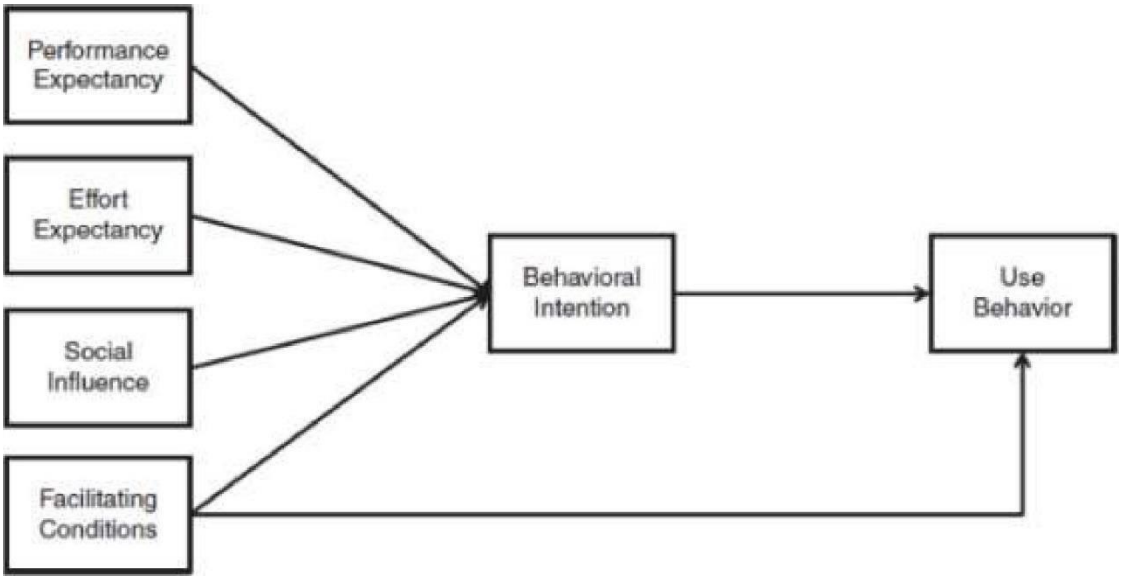


Figure 1. The Unified Theory of Acceptance and Use of Technology (UTAUT) Model (Suki* & Suki, 2018).

The UTAUT model, according to (Suki* & Suki, 2018) served as a foundational framework for understanding the factors that influenced the acceptance and use of the Decision Support System Archiving Hub, encompassed four primary constructs: Performance Expectancy, Effort Expectancy, Social Influence, and Facilitating Conditions. Performance Expectancy represented users' perceptions of the system's effectiveness in enhancing their academic research and management, while Effort Expectancy related to the ease of navigating the system, significantly impacting users' willingness to adopt it.

In the context of NEMSU Tandag Campus, the UTAUT model facilitated a structured exploration of how students and faculty perceived the benefits of the Decision Support System Archiving Hub. By identifying the factors that shaped attitudes and intentions toward the archiving system, the model provided valuable insights for tailoring features and support to promote user engagement.

Through the application of the UTAUT model, this study set the stage for introducing the Decision Support System Archiving Hub, an innovative platform designed to meet the unique needs of academic institutions. Grounded in the principles of U TAUT, this system aimed to be user-friendly, enhancing the accessibility and organization of academic research at NEMSU Tandag Campus.

Conceptual Framework

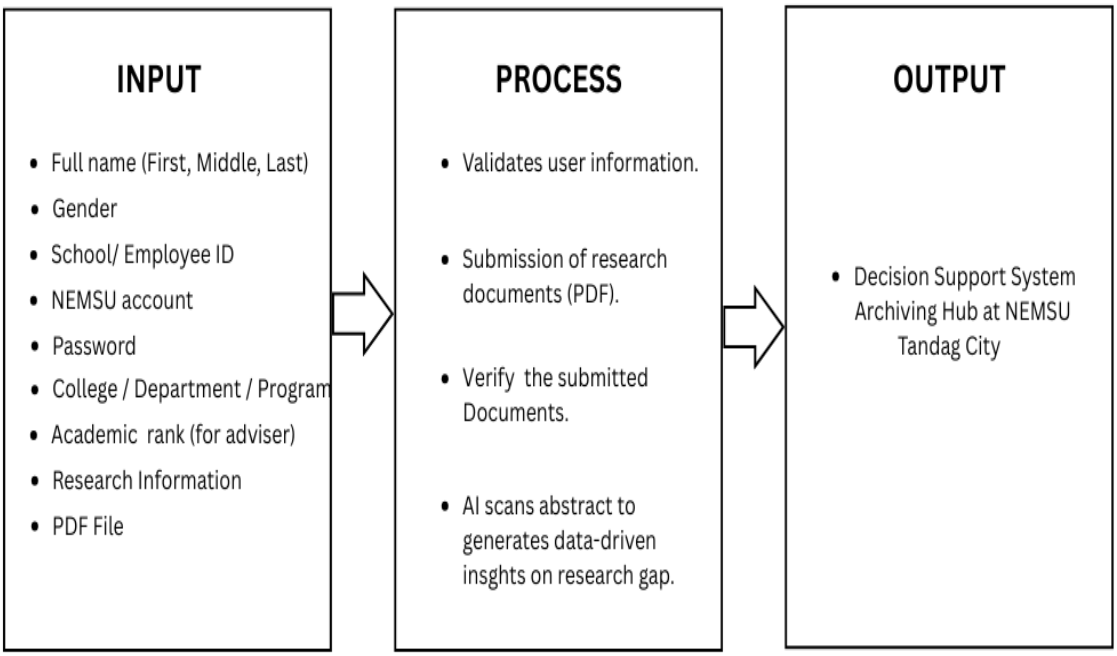


Figure 3. Conceptual Framework of the Study

The conceptual framework for this research illustrates the processing of user and document input to create an intelligent and accessible repository for academic works.

The process begins with the input phase, where essential user and research data are gathered. This includes personal information such as name, email, program, school and employee ID, and password, as well as research specific details like thesis title, keywords, abstract, and supporting documents.

During the process phase, the system validates the user's NEMSU account and securely stores the submitted information. The system integrates Llama to analyze the thesis abstract and identify research gaps. This system process enables intelligent tagging and categorization of academic works, ensuring that each document is systematically archived and organized for easy retrieval.

In the output phase, the system delivers a functional Decision Support System Archiving Hub. This platform provides a centralized, web-based solution for archiving, searching, and retrieving theses and feasibility studies. With its user-friendly interface and AI-powered capabilities, the system streamlines the management of academic documents, enhances search accuracy, and improves access to research materials for students and faculty at NEMSU Tandag City.

The framework demonstrates how user data, AI-driven analysis, and systematic document management interact to create a robust and efficient solution for storing and retrieving academic research.

Statement of the Problem

Within numerous academic institutions, the management of Thesis, and Feasibility Study document is a critical issue. This study seeks to address these issues by investigating potential improvements to the present archiving procedure at NEMSU Tandag Campus. Specifically, the study intended to answer the following questions:

1. What are the limitations of the current archiving system in terms of accessibility, organization, and retrieval of academic documents?

2. How can a system be developed to automate the validation, tagging, and metadata extraction process, while assisting students in identifying research gaps?

3. What digital solutions can be implemented to provide secure, role-based access while supporting efficient archiving and research management?

Objective of the Study

The fundamental goal of this research is to design, implement, and evaluate a website-based system equipped with AI Recommender to streamline the archiving process for thesis documents. To attain this general purpose, the study pursued the specific objectives listed below:

1. Assess the current manual archiving process at NEMSU Tandag Campus, particularly in terms of accessibility, document retrieval, and storage practices.

2. Design and develop a Decision Support System Archiving Hub that:

    - Automates the input of research metadata (title, abstract, author, adviser, etc.) using AI.
    - Detects similarities between submitted studies and existing archives to prevent topic duplication and help users identify related works for research guidance.
    - Identifies possible research gaps from thesis abstracts using LLaMA.

3. Propose and implement a secure, centralized, web-based platform with role-based access for students, instructors, and administrators, allowing efficient and organized management of academic research archives.

Scope and Limitations

This study focused on the development and implementation of a Decision Support System Archiving Hub intended to improve the storage, organization, and retrieval of academic theses and feasibility studies at North Eastern Mindanao State University Main Campus. The system provides a centralized platform where students could browse research titles and AI-generated abstracts, helping them identify existing topics and possible research gaps, while maintaining academic integrity.

To safeguard the content and prevent plagiarism or misuse, only authorized instructors were allowed to upload thesis documents and access their full content. Students were allowed to view the titles and generated gaps while being restricted from uploading files and being limited to viewing content and also to prevent duplicate the existing studies. The system included essential features such as user role management, search and filter options, and categorized

listings to support efficient.

Furthermore, only PDF file formats were accepted for thesis document submissions to ensure standardized. Despite these limitations, the project aimed to deliver an efficient and AI-powered solution for analyzing thesis abstracts to identifying research gaps, ultimately enhancing academic resource management at NEMSU.

Significance of the Study

This study carries significant implications for various stakeholders in the academic research archiving process.

Administrators: Responsible for overseeing the system's operations, ensuring efficient and secure management of academic documents. They handle data organization, storage, and retrieval while monitoring system performance.

Students and Researchers: Gain improved access to theses and feasibility studies, allowing them to identify research gaps, retrieve relevant works faster, and streamline their academic efforts.

Faculty Members: Benefit from the system's data-driven insights for decision making, enabling better research trend analysis and resource allocation.

Academic Institutions: The Decision Support System Archiving Hub enhances document management practices, ensuring reliable, secure, and organized storage of research materials. This promotes a research conducive environment.

Future Researchers: This study serves as a reference for future projects focusing on improving digital archiving, helping researchers refine academic resource management.

Definition of Terms

Archiving Hub: A centralized platform for storing, managing, and retrieving academic research documents, such as theses and feasibility studies. This system allows users to access research abstracts and identify research trends efficiently.

Auto Fill: A feature of the system that uses artificial intelligence to automatically extract and input important information from uploaded thesis or feasibility study documents, such as title, authors, abstract, and adviser. This reduces the workload of manual data input, saves time, and provides accuracy in documentation.

Categorization: The process of organizing documents into specific, defined categories for easier access and retrieval.

Data-driven Insights: Analytical information derived from data, used to enhance decision-making and system functionality.

Data Processing: The process of collecting, organizing, and analyzing data to generate useful insights. In this study, it involves analyzing thesis abstracts to identify research gaps and categorize trends.

DSS (Decision Support System): A computer-based system that helps in decision-making processes by analyzing data and generating actionable insights.

Effort Expectancy: The perceived ease of using the system, which affects how willingly users adopt the platform.

Facilitating Conditions: The availability of resources and support systems that make it easier for users to adopt and use the technology effectively.

Feasibility Study: A detailed analysis that examines the viability of a proposed project or plan.

Llama: An AI-powered large language model (LLM) designed to process and analyze textual data. In this study, Llama is used to process thesis abstracts, identify gaps in research, and suggest trends for academic improvement.

NEMSU (North Eastern Mindanao State University): The institution where the study and system implementation are being conducted.

Performance Expectancy: Users' belief that the system will improve their academic productivity and research capabilities.

Research Gap: Refers to unexplored or underdeveloped areas within a particular field of study. The system identifies research gaps by analyzing the content of thesis abstracts, highlighting areas that need further investigation.

Research Trends: Patterns or directions emerging from a body of research. The system identifies these trends by analyzing multiple thesis abstracts to support future research planning.

Similarity Detection: A tool that is AI-based and utilized within the system to scan newly uploaded studies and match them with previously archived works. This feature aids in detecting similar content even where titles are different,

aiding in avoiding duplication of topics and providing researchers with insights from related studies within the institution.

Social Influence: The degree to which peers, mentors, or other influential individuals affect users' decisions to adopt the system.

Thesis: A document submitted by students to complete their degree, presenting original research on a particular subject.

Thesis Abstract: A brief summary of a research paper, typically including the research problem, objectives, methodology, and findings. The system uses this portion to identify gaps and trends within submitted theses.

Unified Theory of Acceptance and Use of Technology (UTAUT): A theoretical framework that explains user intentions to adopt a technology and their subsequent usage behavior. UTAUT identifies four key constructs: Performance Expectancy, Effort Expectancy, Social Influence, and Facilitating Conditions.

User Interface (UI): The visual elements and layout of a system that allow users to interact with it.

Web-based Platform: A software application accessed and operated through a web browser, allowing users to interact with the system online.

CHAPTER II

Review of Related Literature

In order to have a better idea of the topic and a good start in making the study, the researchers conducted the intensive readings both foreign and local from different resources. These sources included articles and case studies from the internet, and other reading materials. The review of related literature shall served as a foundation to fully understand and meaningfully interpret the results of the ongoing study.

Foreign

The difficulties students encounter when choosing suitable thesis subjects for their courses of study. The authors point out that students may become confused and find it challenging to select appropriate and pertinent themes if the traditional manual method of choosing thesis fields is still used. In order to solve this problem, they created a decision support system that gives students accurate and mathematically supported suggestions for their thesis fields using a profile matching technique. The black box testing approach was used to thoroughly examine this system in order to guarantee its efficacy and dependability. The study's conclusions highlight how decision support tools can help students make better decisions, which will benefit their overall academic experience when developing their theses (Ulsa, A., Sari, M., & Febriyanto, A.,2023).

In order to improve the management of academic publications, particularly theses and graduation projects, their study aimed on building a web-based library system. The system offers features for administrators and students and was built using ASP.NET and a SQL database. Administrators control institutional data, including departments and colleges, while students can browse documents with guest access or contribute research with member access. This system reduces human labor in university libraries and fulfills a requirement for effective, easily accessible academic archiving systems by streamlining document access and storage (Ibrahim, H., & Jadaan, F.,2020).

It highlights the practical benefits of archiving hubs, such as enhancing accessibility, organization, and long-term availability of academic data. Krefft underscores the necessity of structured data management, especially in academic settings where data from research and projects need to be preserved for future use. By centralizing data resources, Decision Support System solutions can improve research workflow and collaboration. The study shows how wellorganized data archives foster more productive academic communities by reducing the effort required to access previous work. This centralization of data also enables efficient retrieval and minimizes redundancy. As a comprehensive solution, Decision Support System serves both students and faculty, helping them locate needed resources quickly. It concludes that digital archiving is a fundamental requirement for academic institutions aiming to optimize data management and research efficacy (Krefft, J.,2021).

The efficient retrieval and utilization of information from meeting recordings and transcripts are crucial in many fields, such as business, academia, and research meetings are important for cooperation, decisionmaking, and information sharing. The recordings and transcripts of meetings are significant resources for future reference and analysis nevertheless, the traditional techniques utilized for accessing and extracting information from meeting recordings can require a significant amount of effort and time. When the user needs to look back on particular information from the meeting recording, they cannot recall where exactly it is located or what exact keyword they have to use to search for it, resulting in difficulties when trying to navigate through extensive amounts of content to find specific insights or debates (Nitin Ghadge, 2024).

As research becomes more and more data intensive, managing this data becomes a major challenge in any organization. At university level there is seldom a unified data management system in place. The general approach to storing data in such environments is to deploy network storage. Each member can store their data organized to their own likings in their dedicated location on the network. Additionally, users tend to store data in distributed manner such as on private devices, portable storage, or public and private repositories. Adding to this complexity, it is common for university departments to have high fluctuation of staff, resulting in major loss of information and data on an employee's departure. A common scenario then is that it is known that certain data has already been created via experiments or simulation. However, it

cannot be retrieved, resulting in a repetition of generation, which is costly and time consuming. Additionally, as of recent years, publishers and funding agencies insist on storing, sharing, and reusing existing research data. We show how digital preservation can help group leaders and their employees cope with these issues, by introducing our own archival system OntoRAIS (Wright et al., 2020).

Local

State that the study assessed the e-library's usability, dependability, performance efficiency, and functional compatibility, among other significant aspects. By providing theses and capstone projects as a valuable resource, the elibrary effectively satisfied the research needs of its users, according to the findings. This encouraging response demonstrates the importance of user centered design in academic systems and implies that technologies can greatly improve the research experience when they are customized to specific user needs. Furthermore, the study underscored the significance of continuous gained which is consistent with the goals of the suggested decision assistance system for choosing thesis subjects in this study (Dela Cruz, J., Albino, R., Moraäa, L., Bachillar, E., & Dullas Jr., R.,2021).

The NIPSC Library's manual thesis and dissertation management inefficiencies and how a contemporary electronic archiving system may be implemented. Librarians and researchers may access academic materials across wired and wireless networks with ease because of the system's N-tier

architecture, which also allows users to read related literature in real time. The addition of SMS functionality improves the user experience even more by giving users instant alerts about material availability. Alpasan's results highlight that the system that was designed guarantees adherence to intellectual property and copyright laws while also increasing the effectiveness of resource management. The importance of technology in changing library services is emphasized by this study, which also emphasizes how academic institutions must embrace digital solutions to improve resource accessibility (Alpasan, F.,2022).

This study explores how online database resources are being used in a private Philippine university, focusing on accessibility, usability, and the impact on research and learning. The findings suggest that access to online databases increases student engagement and supports academic research by offering vast resources beyond traditional libraries. It underscores the importance of digital resources for educational institutions in the Philippines, where limited physical libraries can constrain research. Online databases empower students to perform research from anywhere, fostering a more independent approach to learning. The authors emphasize that digital archives not only enhance research capacity but also promote inclusivity by providing resources to those with limited access to physical libraries. The study concludes that a dedicated digital archiving system would further support research productivity, allowing easier access to past theses and studies. It also mentions that institutions should consider investing in user training to maximize the benefits of these

resources. The study's insights are relevant to digital archiving initiatives aimed at improving academic accessibility (Asio, J. M. R., & Jimenez, E. C.,2020).

The study reveals that digital libraries are particularly beneficial in supplementing classroom materials, providing diverse resources that traditional libraries may lack. Alipio identifies challenges, including limited digital access and resource management skills, that can hinder the effective use of online libraries. However, the study finds that students and faculty who utilize these resources report improved research efficiency and greater exposure to global academic work. The author argues that a more robust, institutionalized digital archiving hub could further strengthen academic efforts, especially in environments where physical resources are limited. Digital libraries also support collaborative learning, as they allow for easy sharing of materials among students and educators. The paper advocates for more support from institutions to maximize digital libraries' benefits, including infrastructure improvements and training. These findings underscore the potential of a Decision Support System archiving hub to enhance access to critical academic resources across Philippine universities (Alipio, M. M., 2021).

The study outlines the structural and technical requirements necessary to implement effective digital archiving, highlighting the need for customized solutions tailored to institutional goals. Gonzales emphasizes that digital archiving systems simplify the storage and retrieval of academic works, particularly for theses, which are essential but often underutilized resources.

The research identifies various challenges, such as funding constraints and technical expertise, which can impact the system's sustainability. Through case studies, Gonzales shows how a structured digital archiving approach improves resource accessibility for students and faculty. The study concludes that digital archiving systems are critical for preserving academic work in Philippine universities, enabling wider resource availability. Gonzales recommends further investment in training and system maintenance to maximize the effectiveness of these digital archiving solutions. This study is highly relevant to the goals of a Decision Support System archiving hub for thesis preservation and access (Gonzales, P. F., 2019).

Synthesis

The literature reviewed, it became clear that digital archiving and decision support systems (DSS) played a crucial role in enhancing the accessibility, organization, and usability of academic resources. These systems helped address common issues such as inefficient data management, difficulty in thesis topic selection, and the challenges of manual resource handling in academic institutions. The studies highlighted how Decision Support System could streamline research processes, reduce redundancy, and support students and faculty in accessing valuable resources more efficiently. Moreover, the adoption of centralized digital archiving systems fostered a more productive research environment, enabling easier access to previous work and contributing to a more collaborative academic community. These insights underscored the potential of a Decision Support System-based archiving hub

to transform thesis and feasibility study management, supporting both academic productivity and long-term resource preservation.

CHAPTER III

METHODOLOGY

This chapter outlined the methodology employed to develop the Archiving Hub. It detailed the research design, system development framework, tools and technologies, data collection techniques, and evaluation methods. This concise overview ensured the reliability of the study and provided a comprehensive understanding of the development and implementation process of the Archiving Hub.

Research Design

Methodical Planning, Risk Analysis, Engineering and Evaluation are necessary for the creation of development NEMSU Archiving Hub. The Spiral Model, similar to the incremental model, places greater emphasis on risk analysis during the system analysis phase. It allows the system to progress through these stages in multiple cycles, ensuring continuous refinement and improvement (Kyeremeh, 2021). Because of its adaptability and iterative nature, which allowed the system to change and grow with each iteration, the Spiral Model was selected for this project.

Figure 4. General Overview of Spiral Model of the Study (Sumaiya Simran, 2024).

Phases of Development

The Spiral Model phases according to the Diagram above are:

Phase I – Planning (Requirement Gathering): In the planning phase, the requirements for the NEMSU Archiving Hub were gathered from stakeholders, including students, faculty and department chair. This involved pinpointing particular difficulties, like the need for effective gap analysis and the difficulty in obtaining resources. To guarantee that the system met user requirements and institutional objectives, the project's scope, objectives, resources, and beginning timetable were established during this phase.

Phase II – Risk Analysis (Risk Management): The risk analysis phase involves figuring out potential risks that might have an impact on the creation and operation of the NEMSU Archiving Hub. These risks could include issues with data privacy, user reluctance to embrace the new system, or technical constraints. To solve these problems, mitigation techniques were developed, including putting strong data security measures in place, holding training sessions, and guaranteeing scalability.

Phase III – Engineering (Designing and Coding Phase): In the engineering phase, the actual development of the NEMSU Archiving Hub took place. This included designing the system architecture, developing core features such as advanced search capabilities and gap identification, and integrating a secure database for archiving.

System Flowchart

This section displays the system flow for the NEMSU landing page archiving hub.

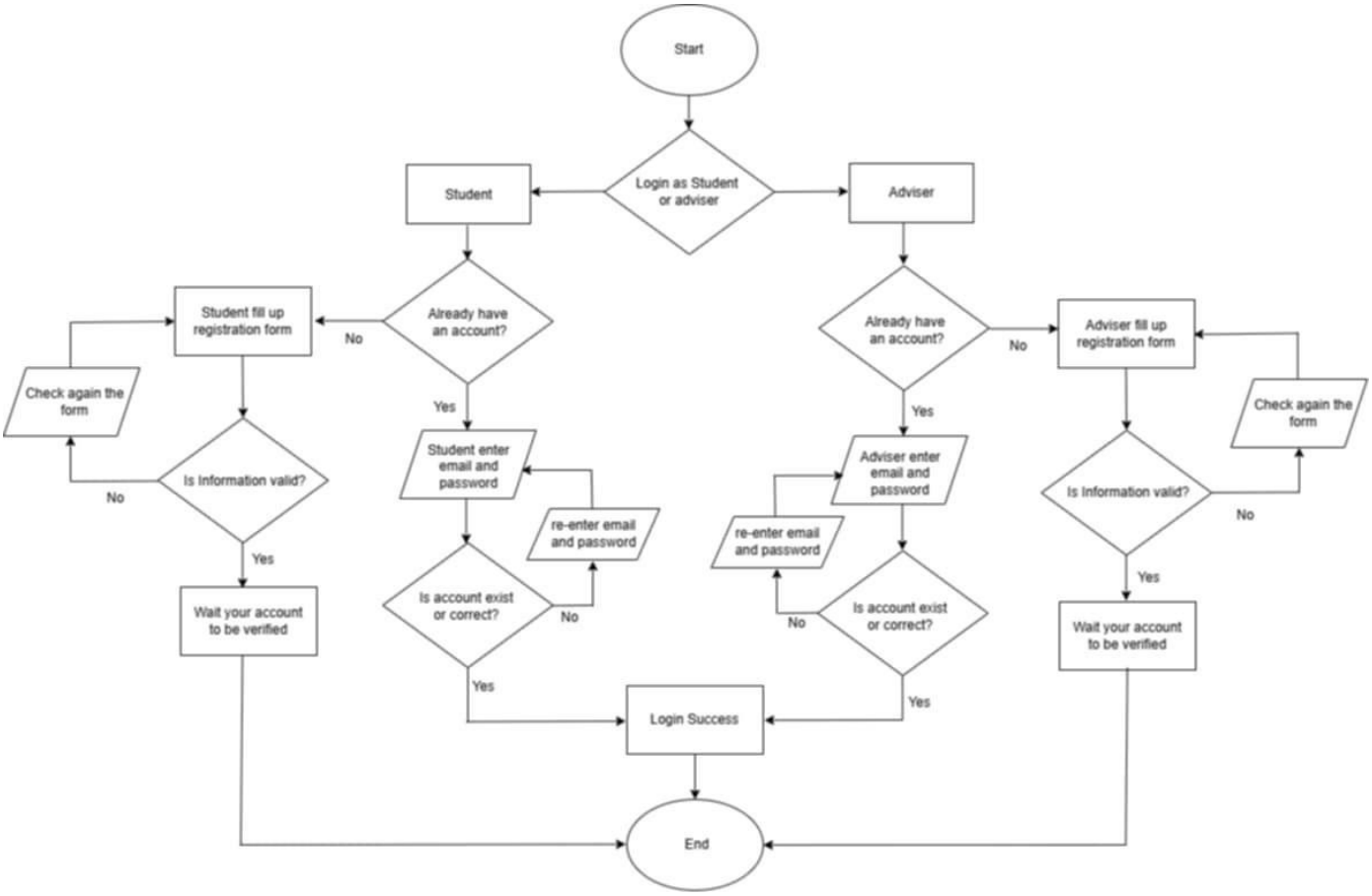


Figure 5. Landing Page of the Archiving Hub

The flowchart describes the user login and registration procedure for students and advisers on the Archiving Hub system's landing page. The users, on reaching the system, select to log in as a student or adviser. If the user already exists, they go ahead and fill in their email address and password. The system checks whether the account is present or not. If the details are correct, the user is given access, and a successful login is accomplished. Otherwise, they are asked to re-enter their credentials.

For new users who do not have an account, the system leads them to complete a registration form. Upon submission, the system verifies the entered data. In case of errors, users are prompted to check and correct the form. If the data is correct, users have to wait for their account to be verified before they can log in.

This process secures a user-specific and secure access system for both students and advisers with data integrity and verification of users as prominent aspects of the access to the system.

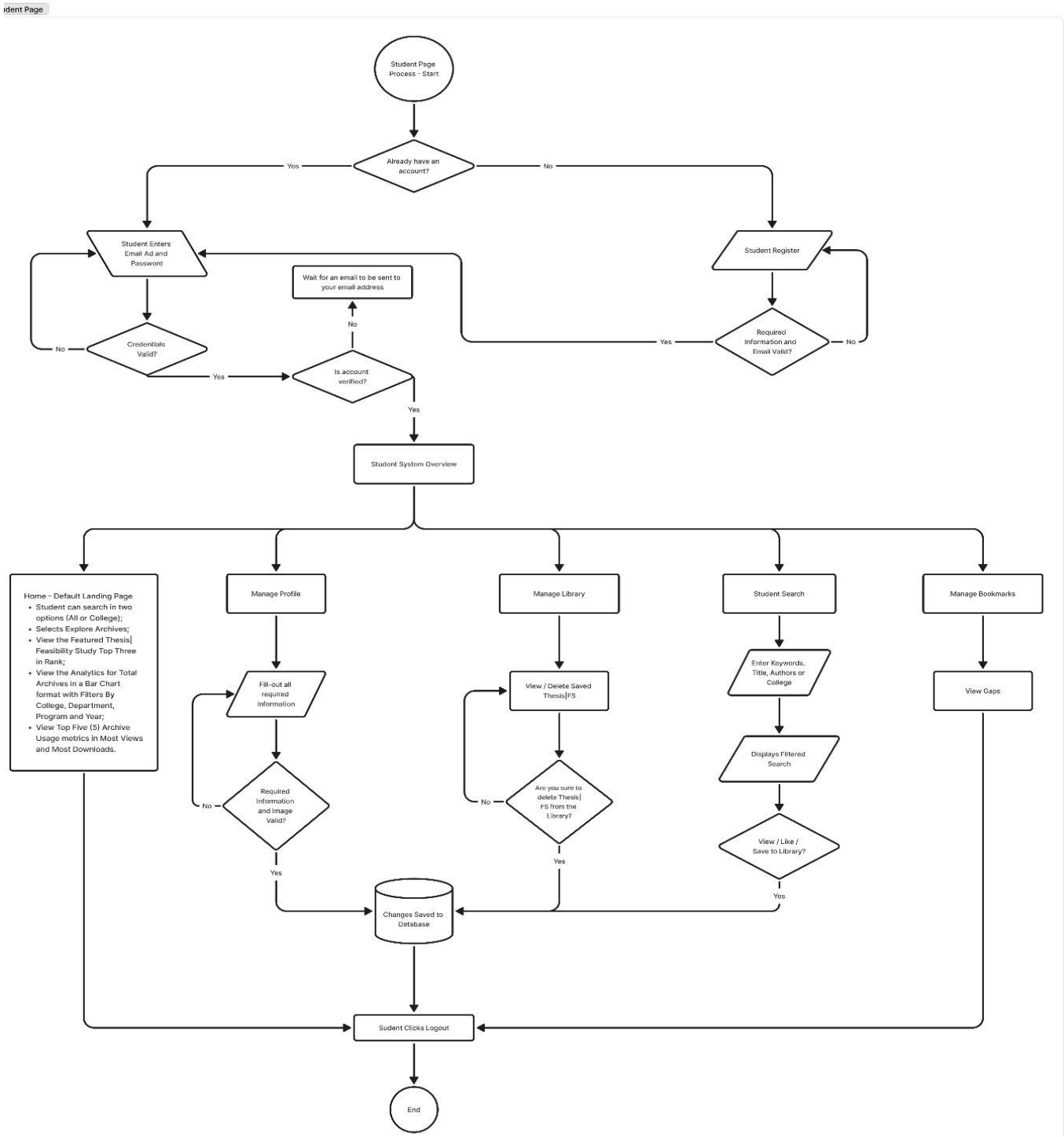


Figure 6. Student Page Flowchart

The flowchart describes the student-side procedure of the Archiving Hub system, starting with the authentication of users. If the student already exists, they enter their password and email. The system verifies the credentials and account verification; if verified, the student is given access to the system. New users are required to register by providing valid data, such as their email, after which they are sent a verification email.

After successful login, the students are routed to the Student System Overview, which serves as the dashboard. They can edit their profile by modifying necessary personal and academic details from there. They can also maintain their personal library, where they can view, delete, and save chosen theses. The student search functionality allows users to search by title, keywords, author, or college and view filtered results which can be viewed, liked, or bookmarked. Students can also bookmark and view research gaps as identified by the AI. The system provides analytics of archived documents with top-viewed, most downloaded, and most liked works by colleges and departments.

All modifications are stored in the database, and users can safely log out upon session termination, providing a comprehensive and organized experience within the archiving hub.

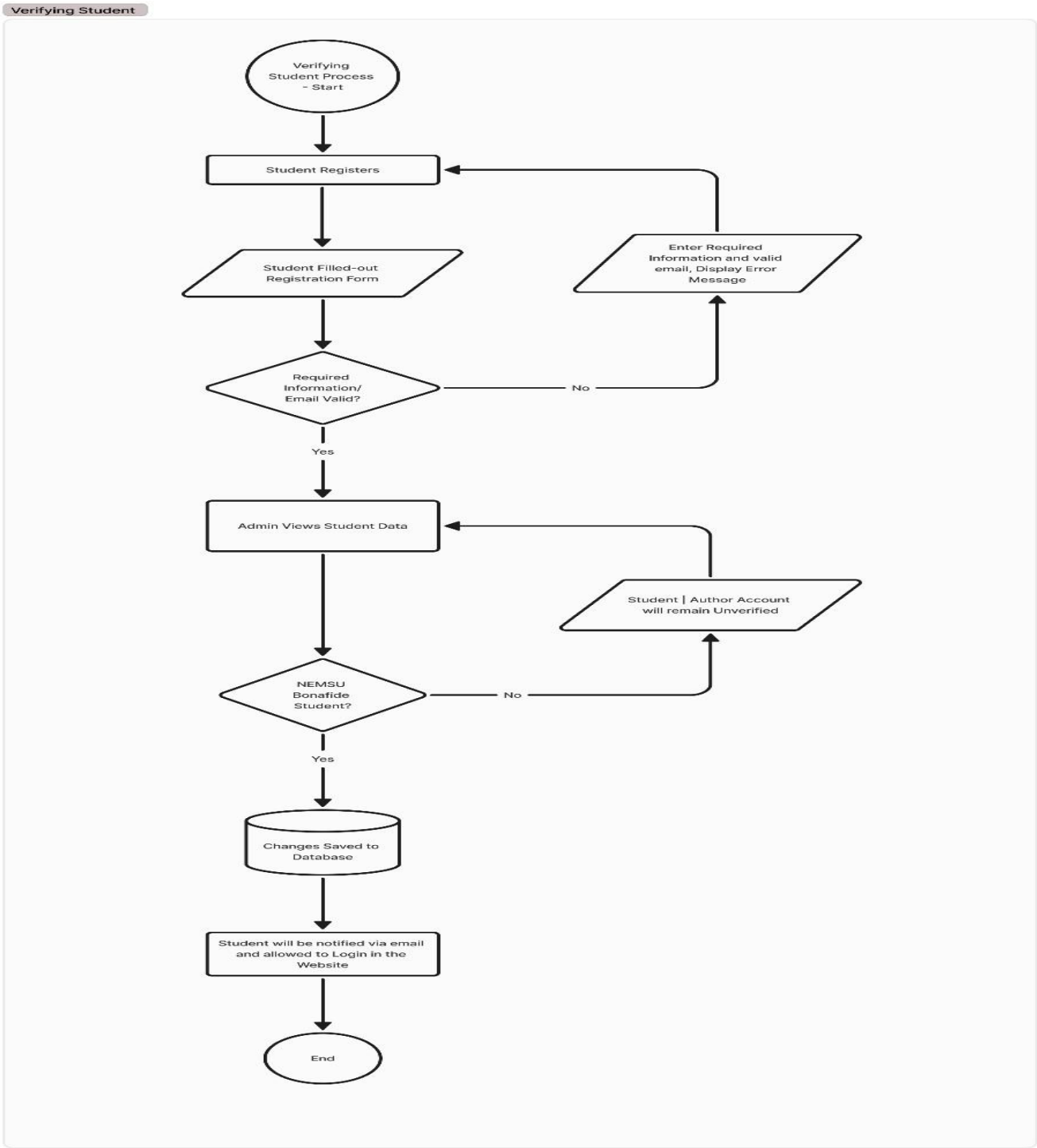


Figure 7. Verification of Student Process Flowchart

The process of student verification starts when a student registers. Once registered, the student completes and submits a registration form with mandatory personal and academic details. The system then confirms if all mandatory fields have been filled and if an email address has been entered. In case any necessary details are lacking or the email is incorrect, the system shows an error message and asks the student to rectify the input and resubmit the form. After the form is filled and submitted correctly, the administrator inspects the student information to authenticate it. Here, the admin decides if the applicant is a genuine student of NEMSU (Northeastern Mindanao State University). If the student is not validated as a legitimate member of the institution, their account is still not verified. But if the student is validated to be legitimate, the changes are committed to the database. The student is then informed through an email that their account has been verified and they are now allowed to log into the website. The process is complete at this stage.

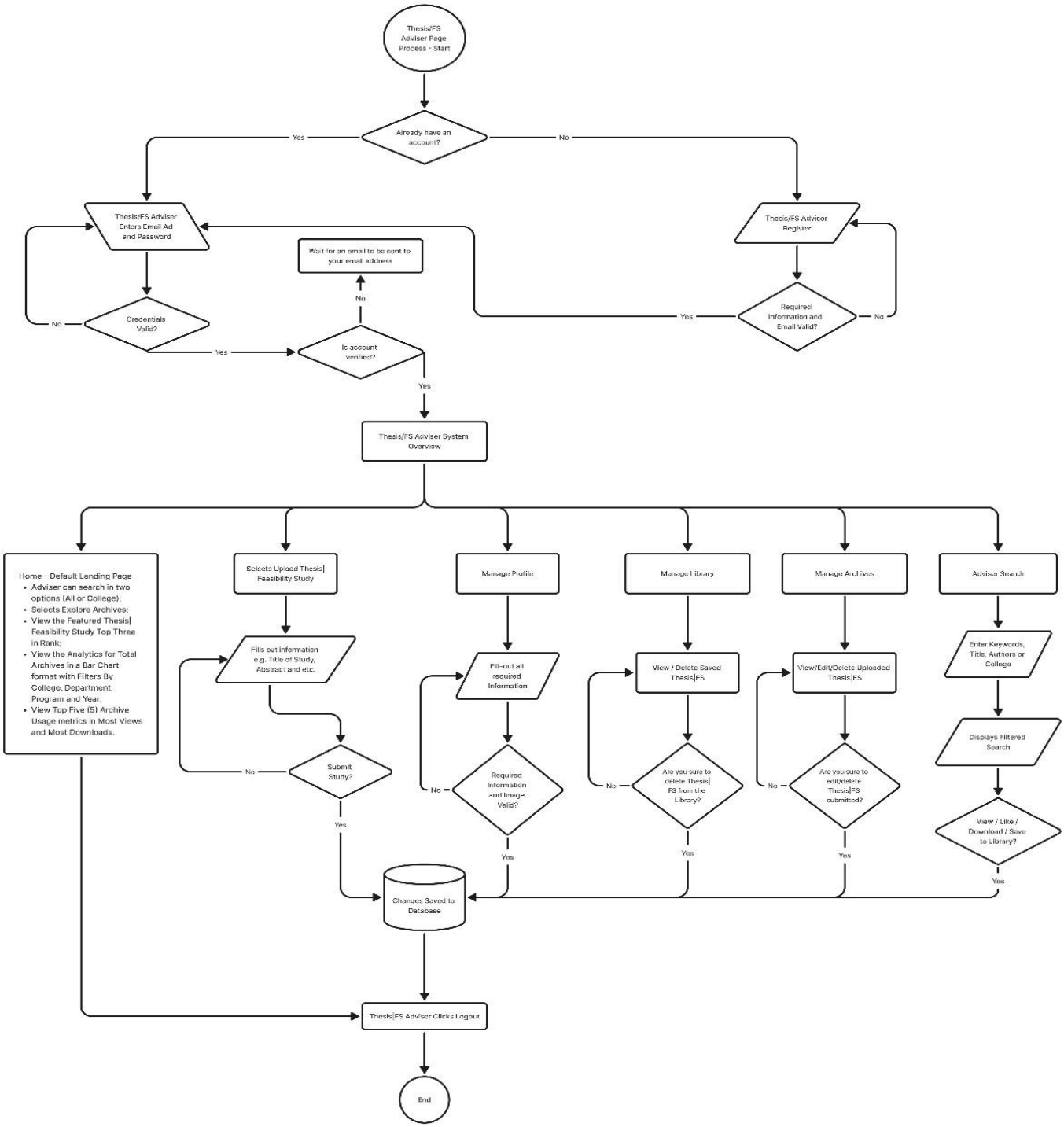


Figure 7. Adviser Page Flowchart

The process starts when a Thesis/FS Adviser logs into the system. If the adviser already possesses an account, they go on to input their email and password. If the credentials are incorrect, they are asked to retry. If correct, and if the account has been verified, the adviser becomes allowed access into the system. If not verified, the system sends an email to finish the verification. For new users, the process initiates with registration. After entering the necessary information and a correct email address, the adviser may proceed to the system after being verified. Once in, the adviser logs in to the Thesis/FS Adviser System Overview, which offers access to a number of features such as the home page, thesis upload, profile management, library management, archives, and search function.

The Home Page shows important statistics and filters to search archived or top thesis/feasibility studies by college, department, and other characteristics. Advisers are able to upload thesis/feasibility studies by providing details like title, abstract, and adviser details. The input is validated before saving into the database. Advisers can also update their profile by changing personal details, which are again validated before saving. In the Library, advisers can see saved theses or search if a particular thesis is already in the system. If not, they are asked to upload. In the Archives section, advisers can check whether to archive a thesis. The Search function provides keyword searches filtered by titles, authors, or colleges, and returns corresponding results which can be seen or downloaded. Once they have finished their work, advisers are free to log out, closing their session.

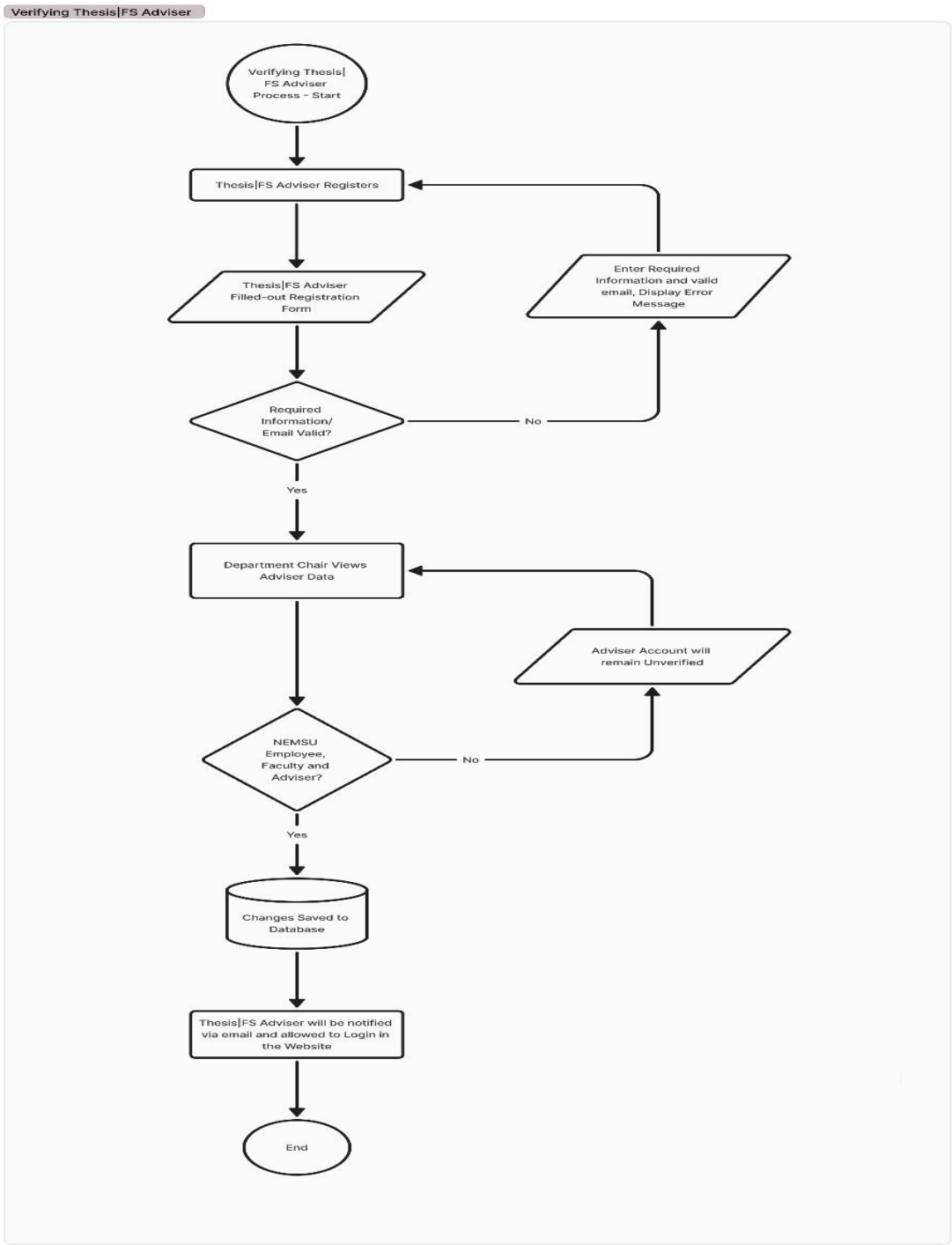


Figure 8. Verification of Adviser Process Flowchart

The Thesis/FS Adviser verification procedure describes the actions to validate and authorize an adviser's registration on an academic platform. The process starts when a thesis or FS Adviser undertakes registration. Upon registration, the adviser completes and submits a registration form containing all necessary details. The platform then verifies whether the form has the necessary details and an authorized email address. If any critical data is incomplete or the email is incorrect, the system asks the adviser to input the correct information and shows an error message. The adviser is then redirected to the registration step again to attempt again.

Once the data is correct, the registration information is sent to the Department Chair, who checks the adviser's information. Here, the decision is taken on whether the adviser is a verified NEMSU employee, faculty member, and authorized thesis/FS adviser. If the person does not qualify under these conditions, the account is kept unverified and the process runs in a loop for rechecking. If the adviser is verified as a bonafide member of the institution, then the data is stored in the database. Lastly, the adviser gets an email notice saying that the verification was successful and can log in to the website. This ends the process.

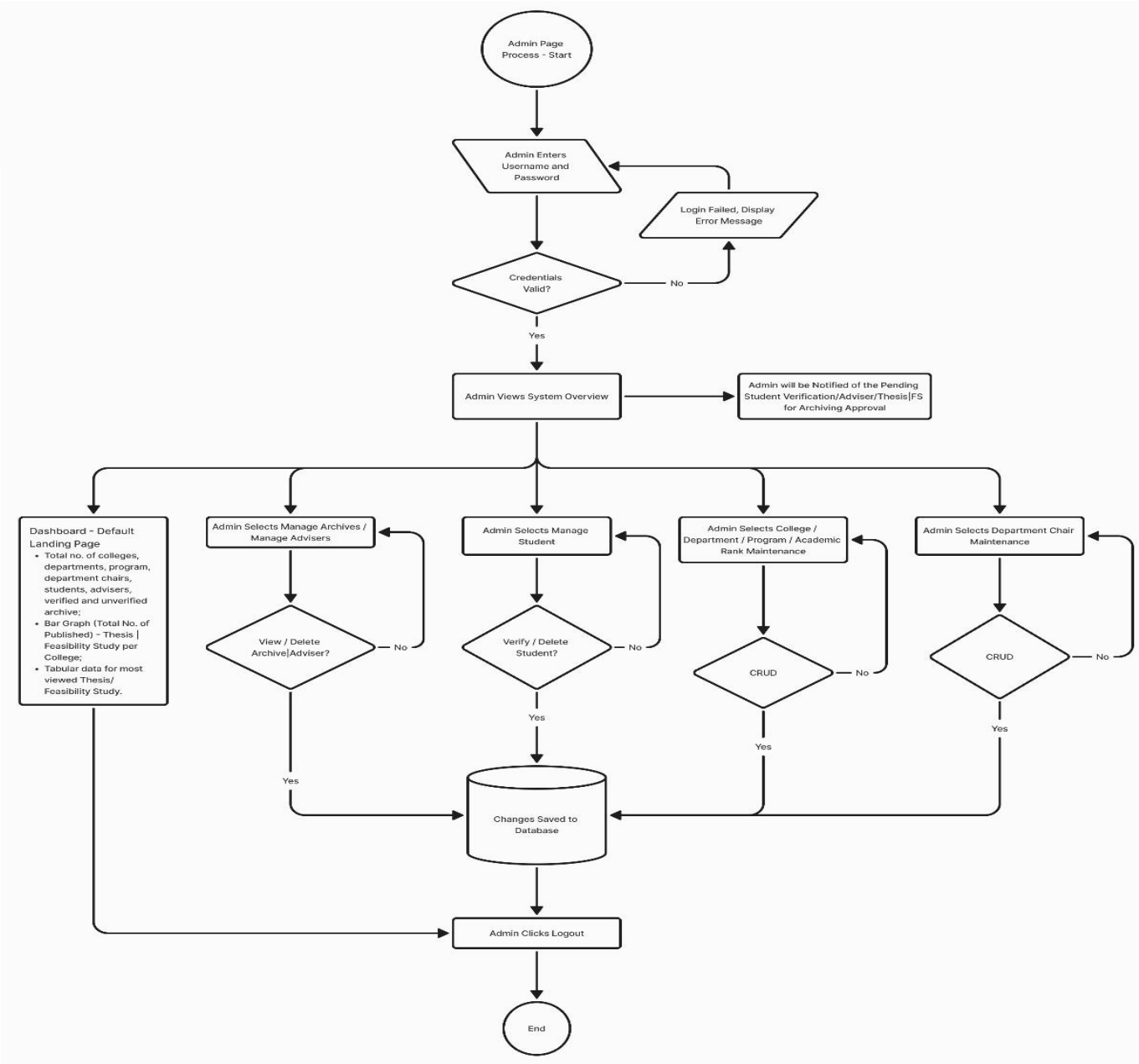


Figure 9. Admin Page Flowchart

The Admin Page Process starts when an administrator inputs his or her username and password to log in to the system. When the credentials are incorrect, the system returns an error message. When logged in successfully, the admin gains access to the system overview and is given a dashboard that shows a summary of important data, such as the number of colleges, departments, program, department chairs, students, advisers, and archived items. It also supports tabular information and bar charts showing analytics like the most popular theses or feasibility studies.

From the overview page, the admin can execute different tasks. They can handle advisers and archived materials by looking at or wiping out records, handle students by checking or wiping out accounts, and keep the institutional organization by executing create, read, update, and delete (CRUD) operations on colleges, departments, programs, academic ranks, and department chairs. If any verification or updates are carried out, the modifications are written directly to the database. Once the required task done, the admin can log out, which closes the session and ends the process.

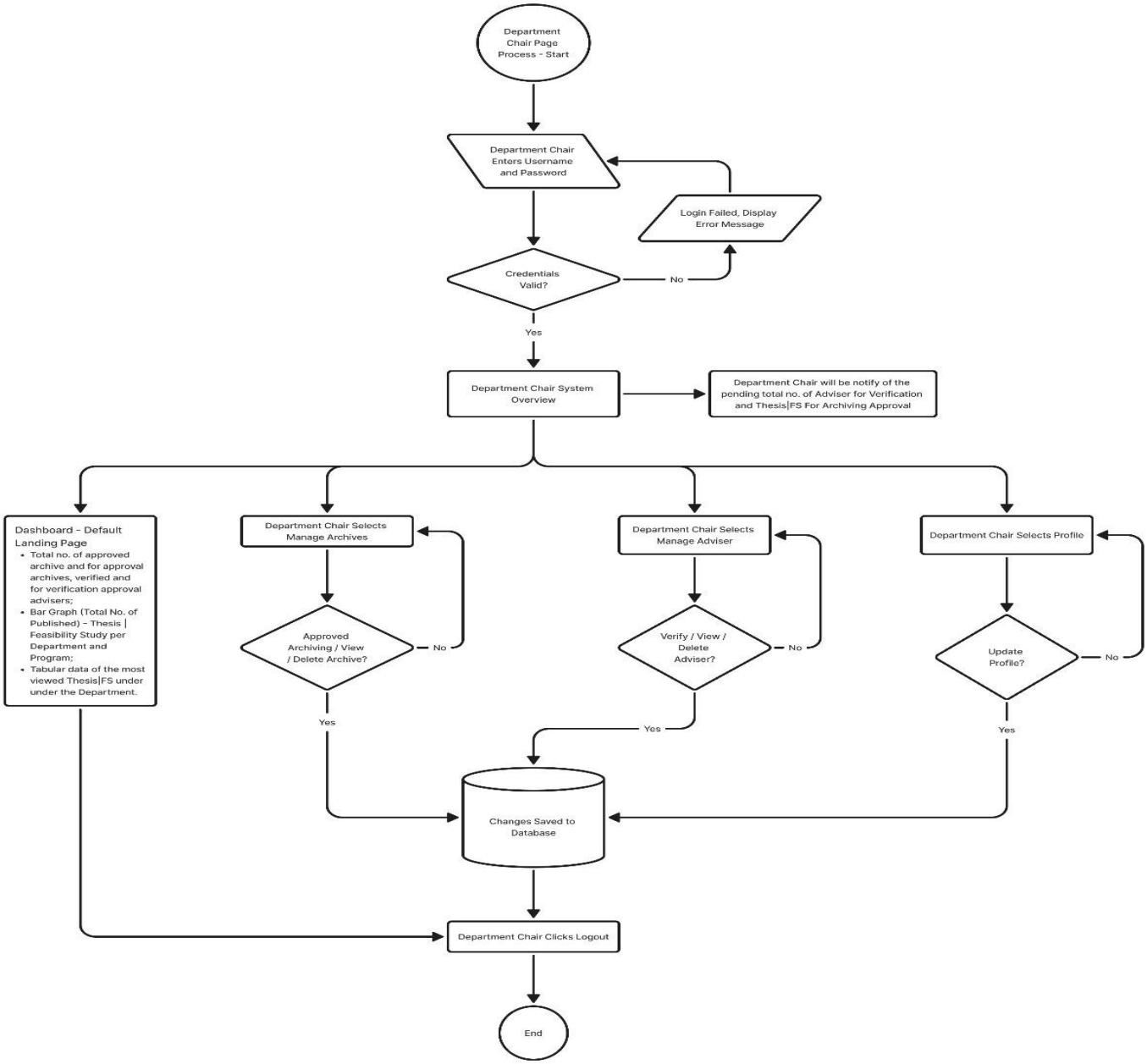


Figure 10. Department Chair Page Flowchart

The Department Chair Page Process starts when the department chair logs into the system by entering their password and username. In case the login details are incorrect, the system will prompt an error message. After logging in successfully, the department chair has access to the system overview, which features a dashboard showing principal information such as approved and pending archives, verified and unverified advisers, and analytics such as bar graphs and tabular information of the most viewed theses and feasibility studies within their department and program.

When entering the system, the department chair receives notification of any outstanding items needing action, including adviser verifications or thesis/feasibility study archiving approvals. From the overview, the department chair can select to manage archives, manage advisers, or update their profile. Managing archives includes approving, viewing, or deleting archive records, while managing advisers involves verifying, viewing, or deleting adviser profiles. If the department head decides to update their profile, the system verifies that any updates were made before saving them. All valid operations result in saving changes in the database. After all operations, the department head logs out, closing the session.

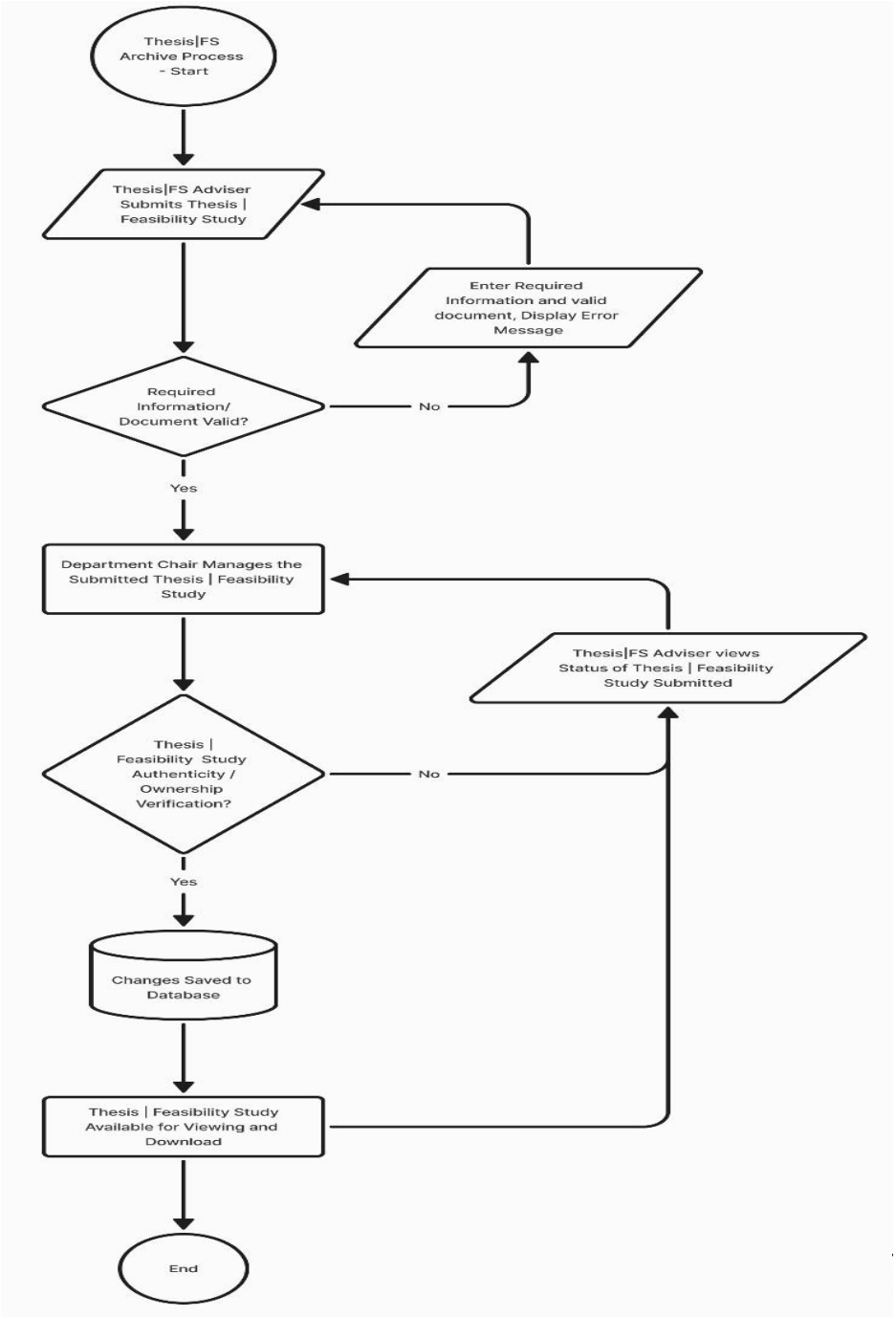


Figure 11. Archive Process Flowchart

The Thesis/Feasibility Study (FS) Archive Process starts when a thesis or FS adviser submits a document for archiving. The system verifies whether all the necessary information and documents are valid. If there is missing information or the document is invalid, an error message is shown, and the adviser has to correct the submission before continuing. After the submission is valid, it is sent to the department chair for management and review.

The adviser assesses the ownership and genuineness of the thesis or FS. When the submission does not pass verification, the adviser can check its status and needs to take remedial action. When the document is verified successfully, the system updates the changes to the database. After verification and storage, the thesis or FS is made accessible for viewing and downloading, indicating the end of the process.

Algorithm Design

The algorithm design of the NEMSU Decision Support System Archiving Hub was centered on integrating artificial intelligence to automate essential academic processes. The system incorporated three core AI-powered features: auto-fill, research gap detection, and similarity detection. All three components were powered by the LLaMA model, which enabled efficient and intelligent processing of academic content. The auto-fill feature utilized LLaMA to extract critical metadata such as the title, authors, adviser, and abstract from uploaded PDF documents. This streamlined data entry, significantly reducing manual workload and minimizing input errors. For research gap detection, LLaMA analyzed the abstract to identify underexplored areas or gaps in the study, providing helpful suggestions for students to refine their research topics. Additionally, the system implemented the K-Nearest Neighbors (KNN) algorithm for similarity detection. By converting abstracts and key content into vector representations, the KNN algorithm calculated the distance between a newly uploaded study and existing archived documents to identify content similarities—even when study titles differed. This feature not only helped prevent topic duplication but also guided students and advisers in discovering related and significant research within the institution. The use of KNN, known for its simplicity and effectiveness in pattern recognition and classification (Altman, 1992), complemented the intelligent capabilities of LLaMA. Collectively, these AI-driven functions transformed the archiving hub into a dynamic and intelligent research management platform.

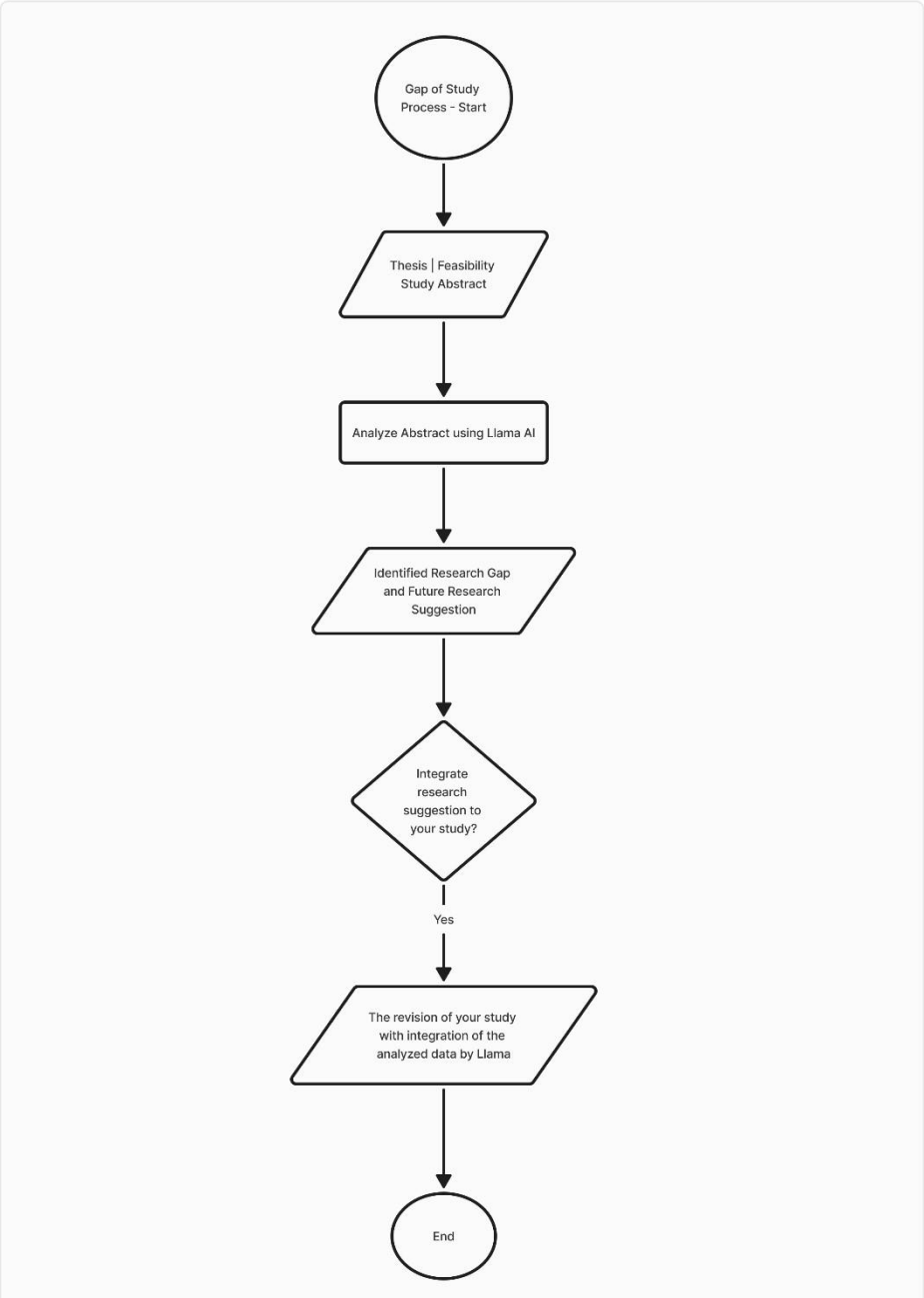


Figure 12. Gap Detection Process Flowchart

The flowchart illustrates the systematic process used in the thesis or feasibility study to identify and address research gaps with the assistance of AI technology, specifically Llama AI. The process begins with the initiation of the study, where the researcher gathers the thesis or feasibility study abstract. This abstract serves as the foundation of the analysis. Once the abstract is ready, it is subjected to analysis using Llama AI, which plays a critical role in extracting meaningful insights and identifying any existing research gaps or areas that warrant further exploration. After the analysis, Llama AI provides suggestions for future research based on the gaps it has detected.

The next stage involves a decision-making process where the researcher evaluates whether or not to incorporate these AI-generated suggestions into their study. If the decision is affirmative, the study undergoes a revision phase. During this stage, the study is enhanced by integrating the data and recommendations derived from Llama AI's analysis. This ensures that the research becomes more comprehensive, current, and aligned with potential areas of academic contribution. The process concludes once the revised study has successfully incorporated the identified gaps and AI insights, signaling the completion of a more refined and valuable research output.

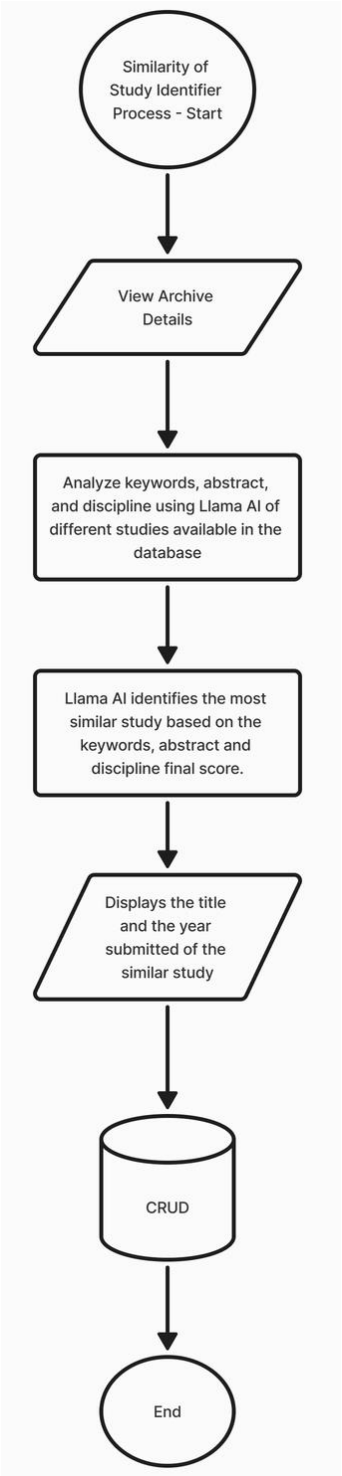


Figure 13: Similarity Detection Process Flowchart

Similarity of Study Identifier Flowchart depicts the procedure of determining similar studies of an academic or research nature with the help of artificial intelligence. The procedure starts with the initiation of the similarity identification process. To start with, the user is asked to see archive information of current studies. Next, the system determines the similarity of various studies archived in the database based on keywords, abstract, and area of study by utilizing Llama AI, which is a machine learning model.

Then, Llama AI calculates a similarity score through the comparison of the input study provided with those in the database on the basis of their keywords, abstract, and discipline classification. It then determines the closest study and goes on to show its title and the year submitted. This data is then processed through a CRUD (Create, Read, Update, Delete) operation, which means that the information can be stored, updated, or read in a database. The process finally terminates.

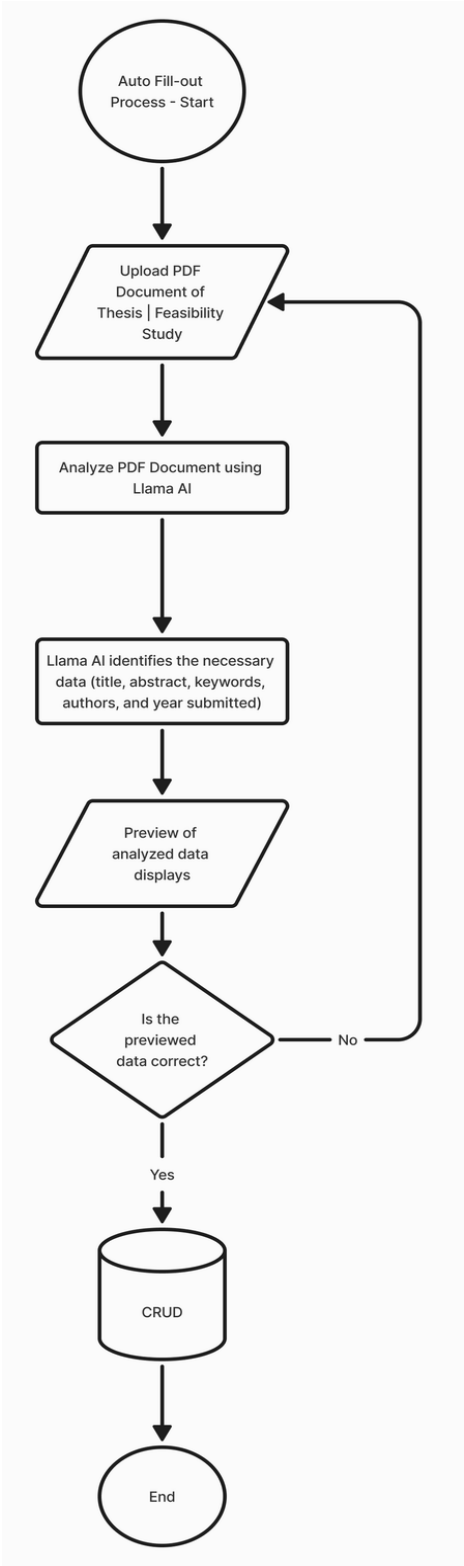


Figure 14: Auto-Fill Process Flowchart

The Auto Fill-out Flowchart explains how to automatically extract and fill in metadata from a thesis or feasibility study PDF document with the use of artificial intelligence. The auto fill-out process starts by initiating the auto fill-out operation. The user is asked to upload a PDF file with the thesis or feasibility study. After upload, the system continues to scan the content of the PDF with the use of Llama AI, an intelligent model with the ability to comprehend academic document formats.

Llama AI then extracts relevant metadata, i.e., the title, abstract, keywords, authors, and the year submitted. This extracted content is then displayed to the user in a preview interface. At this point, the system prompts the user to confirm whether the previewed data is accurate. If the data is inaccurate, the user is prompted to re-upload the PDF to start another analysis cycle. If the data is valid, it gets input into the system for CRUD operations—Create, Read, Update, or Delete—presumably that the data gets stored or gets updated in the database. The process then reaches the end point.

Use Case Diagram

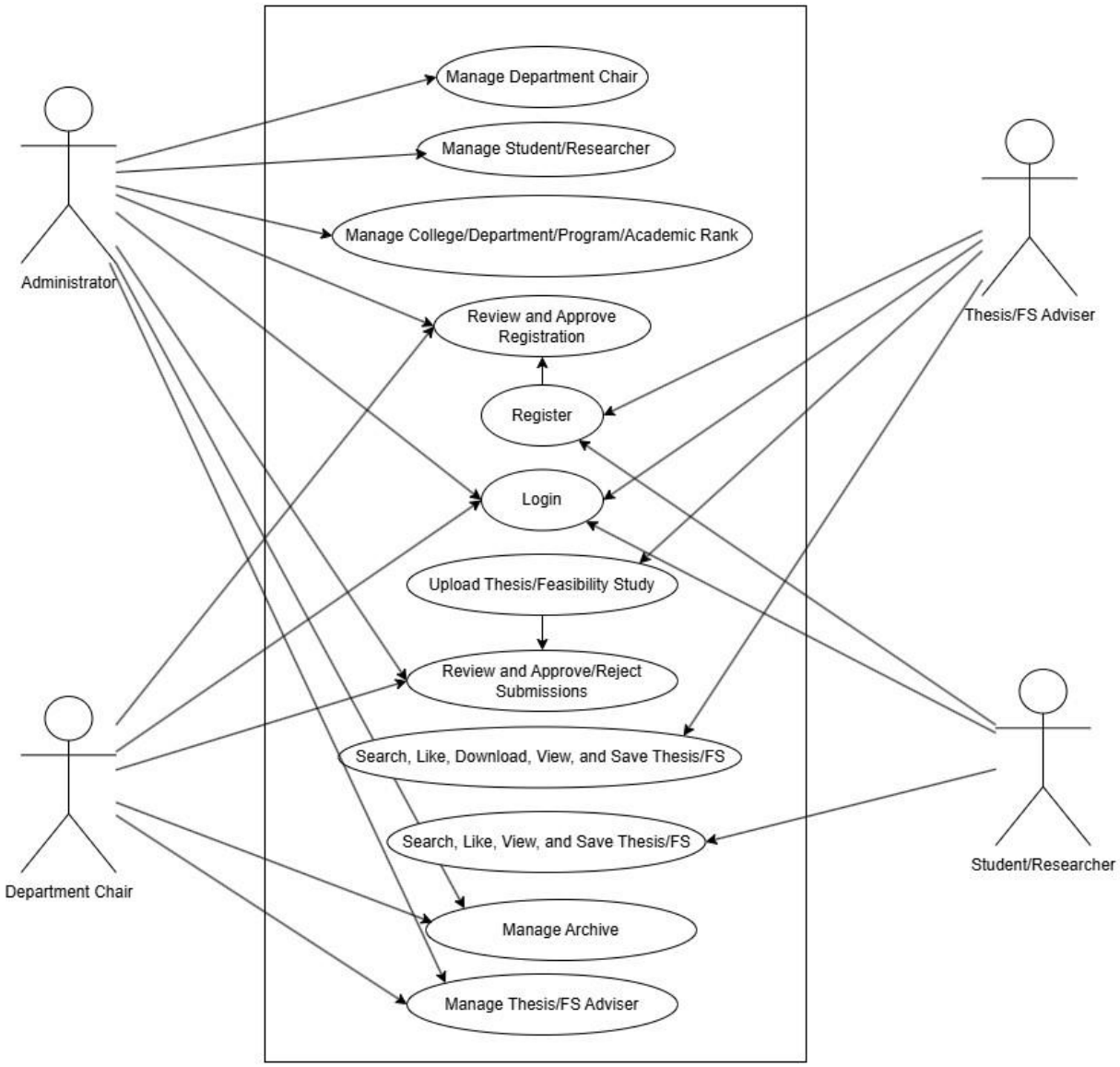


Figure 15: Use Case Diagram

Use case diagram illustrates the primary functions and user interactions for the Thesis/Feasibility Study Archiving System. There are four important users: Administrator, Department Chair, Thesis/FS Adviser, and Student/Researcher. The Administrator has the broadest access and includes department chairs, students or researchers, as well as college-related information such as departments, programs, and academic ranks. Administrators also process user registration approvals, examine thesis or feasibility study submissions, and process archives and adviser data. The Department Chair reviews and processes submitted academic documents, manages archives, and oversees advisers in their department. They also have the ability to view, like, and save thesis or feasibility study documents. The Thesis/FS Adviser submits academic documents for archiving and is engaged in authenticating submissions. They also interact with the system to view or reject works and are able to search and browse archived content. Finally, the Student/Researcher role provides users with registration and login in the system, uploading their thesis or feasibility studies, and searching the repository via searching, viewing, downloading, liking, and saving functionalities. This diagram details how each user helps maintain a working and collaborative digital archive of academic work.

Entity Relationship Diagram (ERD)

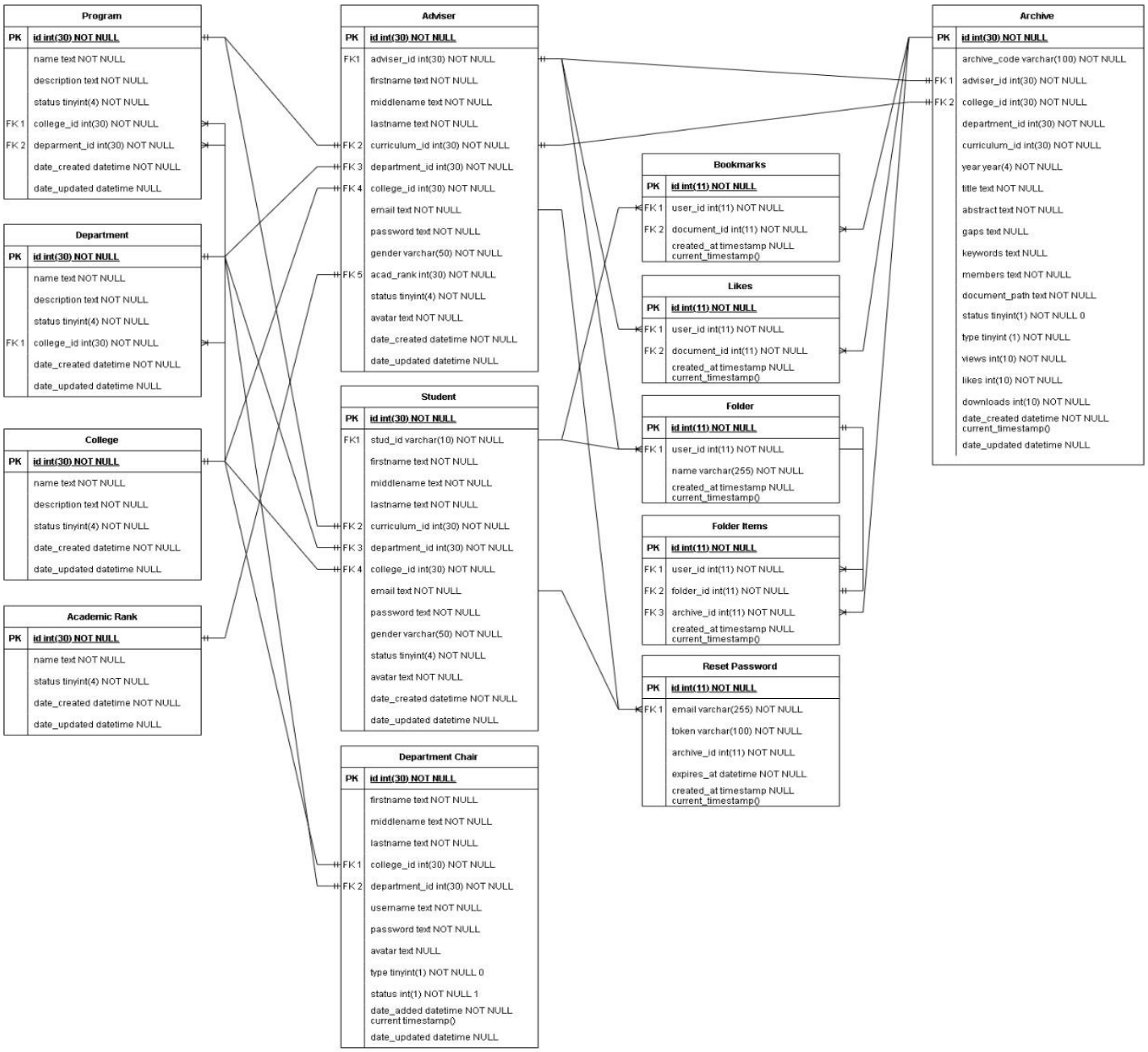


Figure 16. Entity Relationship Diagram (ERD)

The Entity Relationship Diagram (ERD) presents the structure and relationships of the database used in the Thesis/Feasibility Study Archiving System. It consists of some primary entities, namely Adviser, Student, Department Chair, Program, Department, College, Academic Rank, and Archive, and auxiliary entities like Bookmarks, Likes, Folder, Folder Items, and Reset Password. Every entity has attributes that specify its features, e.g., names, IDs, and timestamps for creation and update tracking.

The Adviser, Student, and Department Chair roles are connected to structural roles such as Program, Department, College, and Academic Rank, which assist in classifying users in the academic system. These structural roles make sure that users are connected correctly to their respective academic units.

The Archive entity is at the heart of the system, holding main information regarding submitted theses and feasibility studies, such as metadata such as title, abstract, keywords, views, downloads, and approval status. It refers to users and organizational units for correct attribution and access control. Additional entities such as Bookmarks, Likes, and Folders allow students and researchers to engage with archived documents by saving, liking, or categorizing them.

The Reset Password entity is a utility element that facilitates secure management of user accounts. Generally, the ERD gives a clear picture of how user roles, academic organization, and document management are connected

within the system to maintain data integrity and effective academic resource archiving.

Data Flow Diagram (DFD)

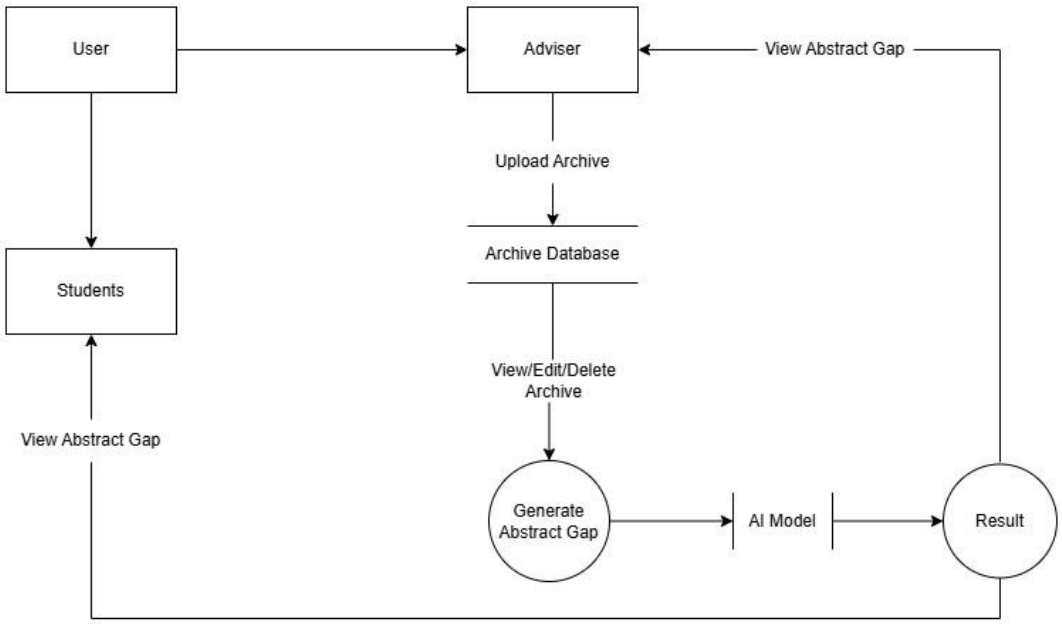


Figure 17. Data Flow Diagram (DFD)

This Data Flow Diagram (DFD) depicts the process flow of Abstract Gap Analysis feature in the Thesis/Feasibility Study Archiving System. The process starts with a User, who may either be a Student or an Adviser, logging into the system. Advisers load archives—theses or feasibility studies—into the Archive Database, where the records are kept. After archiving, the Adviser can view, edit, or delete these documents as required.

The central functionality illustrated is the creation of an abstract gap, a process that assists in the identification of research gaps or untapped areas through the analysis of available abstracts. This activity is triggered from the archive and processed via an AI model, which assesses the content and provides a result indicating possible research gaps. Both Advisers and Students are able to see the abstract gap results, assisting in academic discovery and topic choice. Generally, this diagram highlights the incorporation of AI in the archiving system to maximize research innovation and assist users in finding relevant academic contributions.

Phase IV – Evaluate (Review and Development): The evaluation phase is dedicated to gathering feedback and evaluating the functionality of the system. Students, author, and staff test the functionalities of the NEMSU Archiving Hub system in a controlled environment. Before a system is fully implemented, feedback is utilized to determine what needs to be improved to ensure that users' needs are met.

Research Locale

This study was carried out at North Eastern Mindanao State University (NEMSU) Tandag Campus, selected as the main research site for its alignment with the aims of the A Decision Support System Archiving Hub: Transforming Thesis and Feasibility Study Using Llama. The selection of this site was motivated by the institution's vital role as a repository for academic research outputs, particularly feasibility studies and theses, as well as its growing need for new technology to improve archiving and retrieval processes.

NEMSU functioned as a crucial environment for evaluating and executing the suggested system, owing to the obstacles given by the conventional manual archiving approach, including inefficiency, risks of data loss, and difficulties in retrieving academic materials. Furthermore, the university offered a rich and pertinent context because of the diverse student body and faculty members who were actively engaged in research.

Participants in this study included academics, students, and university employees in charge of research document management, guaranteeing a thorough comprehension of the needs and functionality of the created archiving hub.

Figure 18. Location Map of the Study

Research Respondent

The research of the study engaged a total of 363 participants, which included students from different degree programs at NEMSU Tandag Campus, along with instructors. These participants were subjected to questionnaires, interviews, and system usability testing. Their opinions and comments were crucial in assessing the performance, usability, and feasibility of the recommended Decision Support System (DSS) Archiving Hub of academic papers. Having both students and instructors included in the survey provided a complete view of the system from both user and academic facilitator viewpoints.

Table 1. Distribution and Types of Respondents

Respondents	Population Size	
	Students	**Instructors**
College of Arts and Sciences	115	5
College of Information Technology Education	29	1
College of Teacher Education	47	7
College of Business and Management	153	6
Subtotal	**344**	**19**
Total	**363**	

Data Gathering Procedure

To understand the current manual process of archiving and retrieving academic works at NEMSU Tandag Campus, the research team prepared a formal letter addressed to the department heads, requesting permission to

conduct the study and gather data. Once approval was granted, the research team conducted interviews, faculty members, and selected students to assess their experiences, challenges, and recommendations regarding the existing archiving system.

During the interviews, the researchers presented the proposed DSS Archiving Hub and sought feedback on its features, usability, and potential areas for improvement. The insights gathered were instrumental in tailoring the system to meet the specific needs of administrators, faculty, and students.

Furthermore, the researchers analyzed the current process of archiving thesis and feasibility studies, including submission workflows, categorization methods, and document retrieval practices. These findings provided a comprehensive understanding of the requirements for the proposed system and guided the development of a more efficient and user-friendly digital archiving solution

.

Ethical Considerations

Ethical considerations are fundamental in conducting this study, DSS Archiving Hub: Transforming Thesis and Feasibility Study Management. The researchers ensured open and respectful communication with all participants throughout the data gathering process, which involved interviews, surveys, and system usability testing. Full disclosure was provided to all participants, and

informed consent was obtained prior to their involvement in the study, in compliance with the Data Privacy Act of 2012 (RA 10173).

The study prioritized the safety, confidentiality, and well-being of the respondents, ensuring that all personal and academic information was securely stored and used exclusively for the research purpose. Measures were taken to ensure that no data was shared outside the research context, reflecting a strong commitment to ethical practices in academic research.

The researchers adhered to principles of transparency, credibility, and validity during data collection and system development. Collaboration with participants was conducted in a respectful and ethical manner, ensuring that their feedback contributed to creating a user-centered archiving system. The study's implementation also emphasized fostering a positive relationship with respondents and maintaining trust throughout the research process.

Statistical Treatment

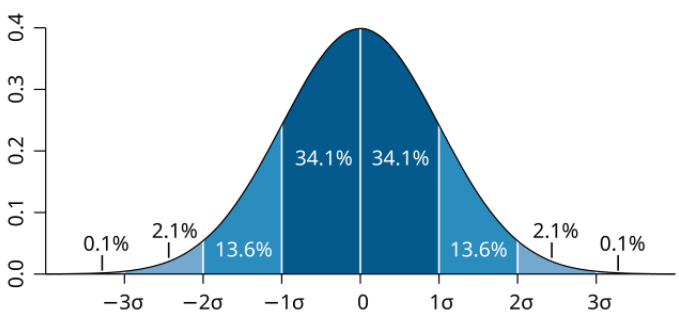
This study utilized a Quantitative Method of Inquiry to analyze and interpret the data gathered. For this approach, the Average Weighted Mean (AWM) was employed to evaluate the respondents' assessments of the system's functionality, usability, and security.

Average Weighted Mean. It provides a comprehensive metric to summarize data and identify trends or patterns in the responses. The formula for the Average Weighted Mean is expressed as:

$$x = \frac{\sum(\boldsymbol{F} \cdot \boldsymbol{X})}{\boldsymbol{N}}$$

The weighted mean (x) is calculated by multiplying the frequency of responses (F) for each item by the weight assigned to that item (X). These products are then summed up, and the total is divided by the total number of responses (N). This gives a weighted average that accounts for both the frequency of responses and the importance or significance of each item based on its assigned weight.

Standard Deviation. A statistical measure that quantifies the amount of variation or dispersion in a set of values. It shows how much the individual data points deviate, on average, from the mean (average) of the data set.



$$s = \sqrt{\frac{\sum (X - \bar{x})^2}{n - 1}}$$

Figure 19. Standard Deviation Graph and Formula

Materials

The development and operation of the DDS Archiving Hub required a combination of software and hardware components. Below were the specific requirements used to ensure the system's seamless functionality and implementation:

Table 2. Software Requirements

Name	Purpose
Windows 10/11	Operating System
Browser	To access the Web Based Platform
PHP	Dynamic Content & Integrated APIs with Database
MySQL	Relational Database Management System
XAMPP	Local Development Environment and Local Web Server
Visual Studio	To access Programming Languages

Table 3. Hardware Requirements

Name	Specification
CPU	2 cores or higher
ROM	128GB SSD or HDD
RAM	8GB
Monitor	Any
Keyboard	Any
Mouse	Any

Schedule and Timeline (Gantt Chart)

The proponents adhered to a schedule and timeline to complete the task, using a work plan and creating a summarized timeline as shown in the table below.

Table 4. Gantt Chart

Activity	Duration Month									
	Aug	Sept	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May
Title Formulation										
Title Defense										
Formulating the Chapter 1										
System Prototyping										
Formulating the Chapters 1 & 2										
Pre-Oral Defense										
Data Collection										
System Development										
Crafting Chapters 4 & 5										
Final Oral Defense Preparation										
Plagiarism/Grammarly Checking										
Deployment										

Project Teams and Responsibilities

Every member of the team has responsibility for the system to be successfully created. The Members have their tasks and collaborate in working together on the pieces of the study. All tasks were given individually to achieve the goal of the study.

Table 5. Team Responsibilities

Proponents	Responsibility
Atriz, Wendel Vencent	• Data Provider • UI Designer • Note-taker
Dedicatoria, Joven	• Data Provider • Encoder • UI Designer
Mendoza, John	• Data Provider • Note-Taker
Planas, Faustine Frichot G.	• Data Provider • Encoder • UI/UX Designer • Document Organizer
Ponciano, Jackelyn	• Data Provider • Encoder • Meeting Minutes Recorder

Budget Cost/Management Plan

The table below shows the budget expenses of this research and system development used as time progresses as well as the cost incurred.

Table 6. Budget Cost Management Plan

Quantity	Unit	Description	Unit Price	Amount
1	App	Laragon	Free	0.00
1	Pdf	Laravel	Free	0.00
2	Sub	Web Hosting	450.00	900.00
1	Unit	Laptop	34,500.00	34,500.00
Total:				**35,400.00**

Chapter IV

PRESENTATION, ANALYSIS, AND INTERPRETATION OF DATA

This chapter covered the tabulation, analysis, and interpretation of the gathered result. The data collected was analyzed using appropriate statistical techniques, and the results are presented in tables to aid in the interpretation and analysis. The chapter concluded with an interpretation of the results and a discussion of the implications of the findings for the research objectives and the field of study. The presentation was the accordance with the sequence of the questions found in the statement of the problem.

Existing Methods in Instructional Material Development

Based on the researchers' problem validation, it was observed that most institutions still relied on manual processes for archiving theses and feasibility studies. These documents were often stored in printed form or kept in scattered digital files without a structured system in place. In many cases, there was no centralized library or repository available for organized storage and access. This made it difficult for students and faculty to retrieve past research, leading to inefficiencies, data loss, and limited reference options. The lack of a proper archiving system also hindered the preservation and visibility of academic outputs, highlighting the need for a more accessible and systematic digital platform.

Proposed System or Platform to Address Instructional Material Challenges

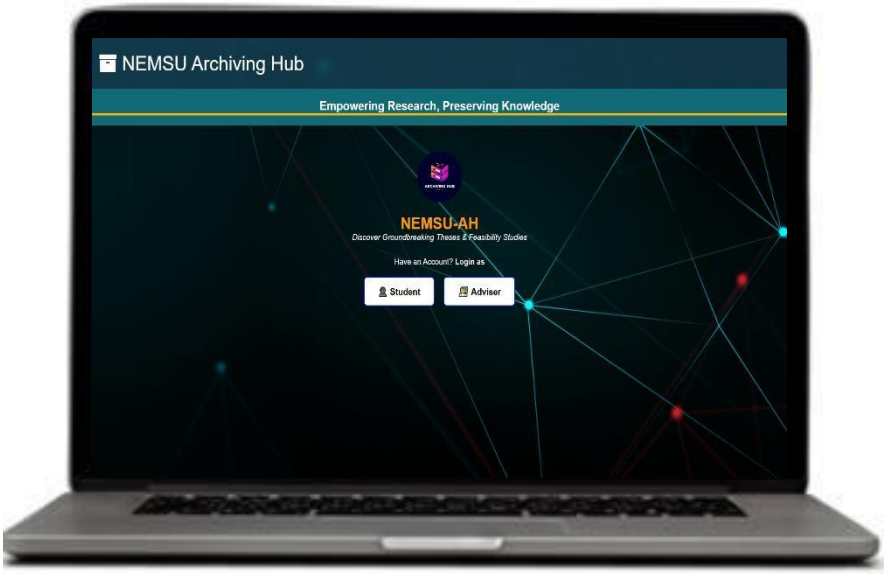


Figure 20. NTSAH landing Page

The researchers proposed the development of Decision Support System Archiving Hub: Transforming Thesis and Feasibility study using Llama designed to address key challenges in research storage, access, and academic integrity within the university. This platform aimed to streamline the archiving of academic work while providing intelligent support to both advisers and students.

For advisers, the Decision Support System Archiving Hub allowed them to upload and access the full content of theses and feasibility studies. This ensured proper documentation and organized storage of research outputs while enabling easy reference and monitoring of submissions across departments. The system also helped advisers in guiding future research by providing a centralized view of completed studies.

For students, the platform offered access to titles, abstracts, and related study information without revealing full content, thereby protecting intellectual property. Students also benefited from an AI-powered feature that analyzed archived studies and generated suggestions for potential research gaps. This helped guide topic selection while ensuring that proposed studies were relevant and non-redundant.

To ensure a fast and efficient response in system operations, the Decision Support System Archiving Hub was developed using native PHP. This choice was driven by PHP's lightweight architecture, direct execution on the server, and robust compatibility with MySQL—factors that allowed for an affordable, smooth, and resource-efficient development process. According to Vayadande, Kuldeep et al. (2024), optimizing object-oriented programming in PHP through techniques such as class design, inheritance, caching, and memory management can significantly improve performance and resource utilization. Compared to heavier frameworks, native PHP enabled developers to optimize memory consumption, minimize server resource usage, and reduce page load time by eliminating unnecessary overhead.

The Decision Support System Archiving Hub thus served as a secure, intelligent, and ethical research environment bridging the gap between academic resource availability and responsible research development.

Challenges and Solutions in Implementing a Decision Support System Archiving hub: Transforming Thesis and Feasibility Study Using Llama.

Implementing the DSS Archiving Hub: An AI-Powered Thesis and Feasibility Study Repository presented several key challenges, along with practical solutions introduced by the researchers. One initial concern was ensuring proper access control between advisers and students. Since students were not permitted to view the full content of archived works to prevent plagiarism, the system was designed to limit student access to titles, abstracts, and AI-generated research gap suggestions only. Advisers, on the other hand, were granted full access to review and assess uploaded works.

Another challenge involved the upload and verification process of research documents. To ensure that only authorized and legitimate submissions were archived, only verified advisers were allowed to upload full copies of research documents. This verification step helped maintain the credibility and integrity of the repository while avoiding the circulation of unofficial or incomplete submissions.

A major concern also surfaced regarding the possibility of topic duplication or plagiarism. To address this, the platform utilized artificial intelligence to scan existing archives and identify similarities or overlaps. When students browsed existing studies, the system automatically suggested research gaps based on previous works. This helped students develop original topics while staying aligned with the university's academic goals.

System access was strictly limited to official users with a verified institutional email address, ensuring that only enrolled students and appointed advisers could interact with the platform. User accounts had to be validated and approved by department heads to maintain a secure and academic-focused environment.

Initially, the system lacked the ability to organize studies efficiently by department, program, or research type. The platform was later improved to support categorized filters and search options, allowing users to browse studies based on academic categories, making it easier to locate relevant works.

Furthermore, to protect intellectual property, the full content of research was hidden from students while allowing them to gather meaningful insights through AI-generated recommendations. This ensured academic integrity while fostering research development.

Evaluation of the level of Efficiency of the Developed System

This section outlined the evaluation of the developed system's efficiency, based on feedback from its users, specifically students and instructors of NEMSU Tandag. A total of 363 respondents took part in the assessment, composed of 344 students and 19 instructors. The evaluation utilized the mean to determine the overall level of agreement or perception regarding each indicator of system efficiency, while the standard deviation was employed to assess the consistency of their responses. These statistical tools together provided an accurate understanding of how users perceived the system's

efficiency and whether their views were generally consistent or showed notable variation.

Based on the Evaluation of the Students

Based on the results gathered from the evaluation conducted among 344 students from various departments of NEMSU Tandag, this section presented an overview of the students' perceptions of the developed system. The evaluation was based on relevant parameters aligned with the ISO 25010 standard. The results provided meaningful insights into the system's strengths as well as areas that might need improvement, ultimately contributing to the enhancement of the students' overall experience in using the platform.

Table 7. Usability Interaction (Interaction Capability) (Students)

Indicators	Mean	Standard Deviation (SD)	Description
1. The system is visually appealing and well-designed.	4.25	0.588	Strongly Agree
2. The navigation is simple and intuitive.	4.18	0.644	Agree
3. I can easily find the features I need.	4.52	0.648	Strongly Agree
4. The system provides helpful guidance and instructions.	4.53	0.596	Strongly Agree
5. The overall user experience is satisfactory.	4.24	0.698	Strongly Agree
Overall	4.34	0.635	Strongly Agree

Usability Interaction in Table 7 showed the assessment of the usability of the system in terms of interaction ability, as rated by student respondents. The system obtained an overall mean score of 4.34 with a standard deviation of 0.635,

which was within the "Strongly Agree" interpretation. This suggested a high degree of satisfaction on the part of student users pertaining to how efficiently and effectively they were able to interact with the platform.

Particularly, students very much agreed that the system provides helpful guidance and instructions 4.53, which suggested that even novice users could use the system with minimal external assistance. The ease of finding required features was also rated highly 4.52, reflecting the user-friendly design and intuitive layout. Visual attractiveness and interface quality were positively recognized 4.25, which ensured a nice experience for the user.

Although the navigation simplicity item had a slightly lower mean 4.18, it was still in the "Agree" category, indicating that although overall easy to use, some slight improvements to navigational flow would enhance interaction further. The user experience was 4.24 overall, supporting that the system satisfied student expectations regarding usability and responsiveness.

Table 8. Functional Suitability (Students)

Indicators	Mean	Standard Deviation (SD)	Description
1. The system provides a complete set of features for uploading, storing, and retrieving academic documents.	4.18	0.630	Agree
2. The system accurately categorizes and organizes thesis and feasibility studies.	4.49	0.687	Strongly Agree
3. Searching and retrieving archived documents is fast and efficient.	4.44	0.659	Strongly Agree
4. The system ensures that all uploaded documents remain intact and accessible.	4.29	0.574	Strongly Agree
5. The system effectively supports academic research and document management.	4.55	0.532	Strongly Agree
Overall	4.39	0.616	Strongly Agree

Functional Sustainability in Table 8 provided the assessment of functional suitability of the system using the responses of student respondents. The system achieved a mean score of 4.39 and a standard deviation of 0.616, which were translated to "Strongly Agree." This meant that most students considered the system functionally complete and responsive to their academic research requirements.

Of the five indicators, the system's effectiveness in supporting academic research and document management had the highest mean score 4.55, which indicated that the students appreciated its utility in the management of the whole research archiving process. Students also greatly agreed that finding and

retrieving documents from the archived ones was speedy and efficient 4.44, and that the platform appropriately categorized and indexed thesis and feasibility studies 4.49. These findings proved that the system allowed users to find information easily, making it convenient for access to research material.

The system also received positive feedback on the integrity and ease of access of uploaded documents 4.29, affirming that stored files were intact and easily retrieved when necessary. While the measure on the completeness of features for uploading, storing, and retrieving documents received a slightly lower mean 4.18, it was still within the "Agree" category, showing that the fundamental functionalities were stable and consistent with user expectations.

Table 9. Reliability (Students)

Indicators	Mean	Standard Deviation (SD)	Description
1. The system ensures high availability with minimal downtime.	4.20	0.485	Strongly Agree
2. The system ensures that archived documents are securely stored and protected from loss.	4.25	0.606	Strongly Agree
3.The system ensures that archived documents remain accessible and protected from accidental loss.	4.61	0.625	Strongly Agree
4. The system operates without unexpected crashes or failures.	4.55	0.642	Strongly Agree
5. The system can recover quickly from errors or downtime.	4.19	0.551	Agree
Overall	4.36	0.582	Strongly Agree

Reliability in Table 9 showed the assessment of the system's reliability by student respondents. The system had a general mean score of 4.36 with a standard deviation of 0.582, which corresponded to "Strongly Agree." This finding reflected that student found the system to be stable and reliable, continually doing its jobs without major intermissions.

Among the distinct measures, the system's capacity to maintain archived documents accessible and secured from unintentional loss gained the highest mean rating 4.61, indicating high confidence in the platform's data integrity features. Similarly, students also had high agreement that the system functioned without unanticipated crashes and failures M = 4.55, pointing to its technical dependability during utilization.

Another highly rated area was the security of saved files, which had a mean rating of 4.25. Students concurred that the system securely stored academic papers and reduced the potential for data loss. The high availability of the system and the low downtime were also noted, as evidenced by the 4.20 mean rating. Yet, the measure of recovery from errors or downtime on the system scored the lowest 4.19, albeit within the "Agree" category. This suggested that although the system was overall reliable, there was still potential for error recovery capabilities to be improved.

Table 10. Performance Efficiency (Students)

Indicators	Mean	Standard Deviation (SD)	Description
1. The system loads and responds quickly when accessing archived documents.	4.32	0.474	Strongly Agree
2.Uploading and downloading files is smooth and does not cause delays.	4.39	0.666	Strongly Agree
3. The system performs well even when multiple users access it at the same time.	4.74	0.467	Strongly Agree
4. The system efficiently handles large amounts of thesis and feasibility study files.	4.69	0.509	Strongly Agree
5. The system optimizes storage space while maintaining document quality.	4.58	0.611	Strongly Agree
Overall	4.54	0.545	Strongly Agree

Performance Efficiency in Table 10 revealed the performance efficiency of the system as assessed through the feedback given by the student respondents. The system garnered a mean score of 4.54 with a standard deviation of 0.545, which was translated as "Strongly Agree." This indicated that most students used the system to a very high degree as being efficient and prompt in response.

Of the five performance measures, the system's capacity to function optimally even when several users were using it at the same time scored the highest mean of 4.74, indicating excellent system stability under simultaneous usage. The same was true for the system's capability to store volumes of thesis

and feasibility study documents 4.69 and to maximize storage space without compromising document quality 4.58. These results proved the system was able to handle user load and file sizes without losing functionality.

The students further concurred that document uploading and downloading were smooth and did not produce any delays 4.39, and that the system loaded and responded rapidly during document access 4.32. These results ascertained the system's dependability in providing fast processing rates and little waiting time, which are essential in supporting learning activities like file submission and retrieval.

Table 11. Perceived Ease of Use (Students)

Indicators	Mean	Standard Deviation (SD)	Description
1. The system is easy to learn, even for first-time users.	4.17	0.565	Agree
2. I can use the system without needing extensive instructions.	4.53	0.615	Strongly Agree
3. The AI-powered features are easy to interact with.	4.24	0.591	Strongly Agree
4. Navigating through different system functionalities is simple and straightforward.	4.59	0.623	Strongly Agree
5. I can quickly complete research-related tasks using the system.	4.25	0.591	Strongly Agree
Overall	4.36	0.597	Strongly Agree

Perceived Ease of Use in Table 11 showed the students' assessment of the system's perceived ease of use. The system got an overall mean rating of 4.36 and a standard deviation of 0.597, which was equivalent to "Strongly Agree." This revealed that students considered the platform easy to learn, understand, and use even without having previous technical experience.

Of the five indicators, the highest mean score of 4.59 was assigned to the statement that it was easy and straightforward to navigate through various system functionalities. This highlighted the intuitive interface of the system, making it easy for users to take actions without getting confused. Moreover, students firmly agreed that they were able to use the system without extensive instructions 4.53 and that the AI-driven features were

easy to use 4.24, which indicated that the automation features were well-integrated and user-friendly.

The participants also concurred that they were able to do research-related activities efficiently with the system 4.25 and that the system was easy to learn, even for new users 4.17. While the last statement had the lowest score among the measures, it was still under the "Agree" category, which meant that the system was still easy to use even by those who were not familiar with digital archiving systems.

Table 12. Security (Students)

Indicators	Mean	Standard Deviation (SD)	Description
1. My personal data is protected.	4.53	0.500	Strongly Agree
2. The system prevents unauthorized access.	4.60	0.514	Strongly Agree
3. The module ensures content integrity.	4.34	0.516	Strongly Agree
4. The system requires secure login credentials.	4.35	0.508	Strongly Agree
5. The AI-generated data is protected from unauthorized modifications through validation and security protocols.	4.23	0.567	Strongly Agree
Overall	4.41	0.521	Strongly Agree

Security in Table 12 outlined the system's security assessment from the students' feedback. The system had a general mean score of 4.41 with a standard deviation of 0.521, interpreted as "Strongly Agree." This was an indication of high confidence from the students in the ability of the platform to secure academic and personal information.

The most highly rated of these, with an average rating of 4.60, was the system's access control, which showed that students highly valued the level of access control in the system. In second place came the belief that personal information was secure 4.53, which showed evidence of high trust in the data privacy of the system.

The use of safe login credentials by the system 4.35 and how it could ensure content integrity 4.34 also met with strong agreement, indicating that students believed their academic work was safe and unaltered. Furthermore, the

protection of AI-based data from any unauthorized alterations through validation and security features met with a somewhat lower, but still positive, rating of 4.23, indicating overall approval with some degree of variation in perception.

The relatively low standard deviations for all the indicators, between 0.500 and 0.567, reflected uniform responses from the students. The uniformity testified to the reliability of the feedback and suggested collective confidence in the security functions of the system.

Evaluation of the Instructors

Based on the results of the evaluation conducted among 19 instructors across different departments and colleges of NEMSU Tandag, this section provided an overview of the instructors' perceptions. The evaluation covered the same parameters as the students' evaluation. The findings offered valuable insights into areas of strength and potential areas for improvement, aiming to enhance the overall experience of using the platform. The results were presented in separate tabulations, as instructors had different experiences and interactions with the platform compared to students.

Table 13. Usability Interaction (Interaction Capability) (Instructors)

Indicators	Mean	Standard Deviation (SD)	Description
1. The system is visually appealing and well-designed.	3.79	1.357	Agree
2. The navigation is simple and intuitive.	4.47	0.513	Strongly Agree
3. I can easily find the features I need.	4.21	0.855	Strongly Agree
4. The system provides helpful guidance and instructions.	4.26	1.046	Strongly Agree
5. The overall user experience is satisfactory.	4.16	0.898	Agree
Overall	4.18	0.934	Strongly Agree

Usability Interaction in Table 13 showed the instructors' ratings of the usability of the system with regard to interaction capability. On the basis of the responses of 19 instructors from different departments of NEMSU Tandag Campus, the system had an overall mean rating of 4.18 with a standard deviation of 0.934, which meant "Strongly Agree." This indicated that instructors in general considered the system to be user-friendly and interactive, though some deviation in their experiences was noted.

Among the specific measures, navigation of the system had the highest mean rating of 4.47, with a standard deviation as low as 0.513, which meant that instructors most agreed on the clarity and simplicity of the navigation structure of the platform. This was reflective of an effective system interface that allowed users to navigate through its capabilities with minimal effort. The ease

with which required features could be located also received near-universal support, at a mean of 4.21 and a standard deviation of 0.855, indicating a well-arranged and coherent layout, although there were some variations in users' perception.

The effectiveness of the system's instructions and guidance was also positively recognized, with a mean of 4.26. Yet, the comparatively higher standard deviation of 1.046 suggested that although a majority of instructors found the instructions effective, there could have been variability in how clear or complete they seemed to various users. Likewise, the overall experience of the users rated a mean of 4.16 (SD = 0.898), reflecting overall satisfaction, though certain instructors might have faced some minor usability issues.

Notably, the lowest rating was obtained for visual design of the system, where the mean was 3.79 and the high standard deviation of 1.357 resulted. Even though teachers were unanimous in finding the system acceptable in terms of visual appearance, the large variation of responses indicated varying levels of opinion on its niceness or quality of interface design.

Table 14. Functional Suitability (Instructors)

Indicators	Mean	Standard Deviation (SD)	Description
1. The system provides a complete set of features for uploading, storing, and retrieving academic documents.	4.42	0.769	Strongly Agree
2. The system accurately categorizes and organizes thesis and feasibility studies.	4.32	0.582	Strongly Agree
3. Searching and retrieving archived documents is fast and efficient.	4.21	0.918	Strongly Agree
4. The system ensures that all uploaded documents remain intact and accessible.	4.37	0.597	Strongly Agree
5. The system effectively supports academic research and document management.	4.53	0.697	Strongly Agree
Overall	4.37	0.713	Strongly Agree

Table 14 presented the assessment of the functional suitability of the system from the answers of 19 instructors from different colleges and departments of NEMSU Tandag Campus. The findings pointed to the system being operationally sound and academic requirement-compatible, as reflected through a mean of 4.37 and a standard deviation of 0.713, both within the "Strongly Agree" interpretation range. This meant that the instructors rated the system high in terms of its effectiveness in providing its target functionalities, specifically in managing academic documents like thesis and feasibility study.

Out of the five indicators measured, the highest score went to the statement, "The system effectively supports academic research and document

management," with a mean score of 4.53. This revealed that instructors significantly recognized the role of the system in supporting research activities, ranging from storage to retrieval and arrangement of scholarly work. Likewise, the system's ability to preserve the integrity and accessibility of uploaded documents was also ranked quite high, with a mean of 4.37, testifying to instructors' trust in the reliability and data management capabilities of the system.

In addition, the feature facilitating effective searching and retrieval of archived documents scored a positive mean of 4.21, indicating that the system's retrieval functions were, on average, quite effective, albeit potentially with some minimal scope for improvement. The system's capability to correctly categorize and sort scholarly work also received a robust score of 4.32, further upholding the evidence that the site offered systematic and organized access to research products. Finally, feature completeness in document uploading, storing, and fetching received a mean of 4.42, highlighting the depth of the system's functional mechanisms.

Table 15. Reliability (Instructors)

Indicators	Mean	Standard Deviation (SD)	Description
1. The system ensures high availability with minimal downtime.	4.37	0.684	Strongly Agree
2. The system ensures that archived documents are securely stored and protected from loss.	4.37	0.684	Strongly Agree
3. The system ensures that archived documents remain accessible and protected from accidental loss.	4.47	0.697	Strongly Agree
4. The system operates without unexpected crashes or failures.	4.21	0.918	Strongly Agree
5. The system can recover quickly from errors or downtime.	4.05	0.848	Agree
Overall	4.29	0.766	Strongly Agree

Table 15 presents the instructors' evaluation of the system's reliability. The measure revealed that generally the Decision Support System Archiving Hub was deemed to be reliable as seen with the overall mean value of 4.29 and standard deviation of 0.766 translating to the measurement "Strongly Agree." The reading meant the system worked correctly on all instances and reduced distractions in the form of crashes or data loss.

Among the distinct indicators, two items had the same mean rating of 4.37, which were: "The system ensures high availability with minimal downtime" and "The system ensures that archived documents are securely stored and protected from loss." These ratings indicated that teachers had faith in the system to be available when needed and to keep academic material secure

from threats. The most highly rated indicator was "The system ensures that archived documents remain accessible and protected from accidental loss," at 4.47. This was an indication of instructors' very high belief in the platform's security measures for data continuity and integrity.

Nonetheless, the affirmation "The system runs without crashes or failures" was scored lower at 4.21, and "The system recovers quickly from errors or downtime" had the lowest mean of 4.05, which was translated as "Agree." These findings indicated that although the system was normally stable, there could be rare problems concerning crashes or slow recovery during technical disruptions. However, the general high ratings in all measures of reliability indicated that teachers perceived the system as a reliable means of storing and retrieving academic papers with little interference.

Table 16. Performance Efficiency (Instructors)

Indicators	Mean	Standard Deviation (SD)	Description
1. The system loads and responds quickly when accessing archived documents.	4.21	1.032	Strongly Agree
2. Uploading and downloading files is smooth and does not cause delays.	4.37	0.761	Strongly Agree
3. The system performs well even when multiple users access it at the same time.	4.58	0.692	Strongly Agree
4. The system efficiently handles large amounts of thesis and feasibility study files.	4.26	0.733	Strongly Agree
5. The system optimizes storage space while maintaining document quality.	4.53	0.612	Strongly Agree
Overall	4.39	0.766	Strongly Agree

Table 16 showed the instructors' assessment of the performance efficiency of the system. The system achieved a total mean of 4.39 and a standard deviation of 0.766, which was measured as "Strongly Agree." This finding indicated that the system operated efficiently with quick and responsive operations when used. NEMSU Tandag Campus instructors concurred that the system performed tasks with minimal delay and could effectively perform operations even under heavier usage.

Of the five performance indicators, the most highly rated was "The system functions well even under simultaneous access by many users," with a mean of 4.58 and a standard deviation of 0.692. This reflected very high confidence that the system could both stay stable and responsive under

concurrent access, which was critical in an academic environment where several users might be downloading or uploading documents simultaneously. The second most highly rated option was "The system optimizes storage space without losing document quality," which was given a mean of 4.53, highlighting the platform's ability to optimize storage space without losing the quality of scholarly files.

File uploading and downloading were also positively rated, with a mean rating of 4.37, indicating that the system supported seamless file uploading and downloading that did not give rise to serious delays. In addition, the teachers concurred that "The system effectively processes large volumes of thesis and feasibility study documents," with a mean rating of 4.26, indicating that the system was capable of accommodating heavy academic paper loads. Finally, the capability of the system to "load and respond very fast when accessing archived documents" was rated 4.21, which while the lowest of the indicators, was still under the "Strongly Agree" category.

Table 17. Perceived Ease of Use (Instructors)

Indicators	Mean	Standard Deviation (SD)	Description
1. The system is easy to learn, even for first-time users.	4.53	0.612	Strongly Agree
2. I can use the system without needing extensive instructions.	4.21	0.631	Strongly Agree
3. The AI-powered features are easy to interact with.	4.47	0.612	Strongly Agree
4. Navigating through different system functionalities is simple and straightforward.	4.32	0.671	Strongly Agree
5. I can quickly complete research-related tasks using the system.	4.37	0.761	Strongly Agree
Overall	4.38	0.657	Strongly Agree

Table 17 showed the instructors' rating of the system's perceived ease of use, which resulted in a total mean score of 4.38 and a standard deviation of 0.657, both of which were translated as "Strongly Agree." This meant that instructors considered the system very user-friendly, easy to learn, and easy to use even with little technical guidance. Of the personal indicators, the highest-rated was "The system is easy to learn, even for first-time users," with a mean of 4.53, indicating a strong consensus that the system didn't need much or any training to use effectively. This confirmed the platform's ease of use, especially for those who had not used digital archiving systems before.

Further, the assertion "The AI-enriched features are easy to work with" had a very high mean value of 4.47, indicating that instructors found it

simple and not intimidating to work with the integration of AI technology. Likewise, "I could complete research-related work quickly using the system" achieved a high value of 4.37, highlighting the effectiveness of the system in streamlining academic processes. Ease of movement through the system's different capabilities was also rated highly, at 4.32, again showing that the interface had been found easy and well-organized. Finally, the declaration "I was able to use the system without needing comprehensive instructions" received an average of 4.21, which, as the lowest of the indicators, was still in the "Strongly Agree" category, and indicated that although a little instruction would be useful, the system as a whole was self-explanatory.

Table 18. Security (Instructors)

Indicators	Mean	Standard Deviation (SD)	Description
1. My personal data is protected.	4.37	0.831	Strongly Agree
2. The system prevents unauthorized access.	4.37	0.684	Strongly Agree
3. The module ensures content integrity.	4.26	0.872	Strongly Agree
4. The system requires secure login credentials.	4.47	0.612	Strongly Agree
5. The AI-generated data is protected from unauthorized modifications through validation and security protocols.	4.42	0.607	Strongly Agree
Overall	4.38	0.721	Strongly Agree

Table 18 showed the instructors' assessment of the system's security features, which pointed to its efficiency in guarding sensitive information and ensuring the integrity of stored documents. The system attained an overall mean score of 4.38 and a standard deviation of 0.721, which corresponded to "Strongly Agree." This finding indicated that the instructors trusted the platform's security systems significantly. Among the particular statements, the most highly rated one was "The system required secure login credentials" with a mean of 4.47, followed by "The AI-generated data was protected from unauthorized modification through validation and security measures" at 4.42. The results highlighted that instructor appreciated the platform's strong authentication and data security measures.

Both the assertions "My personal data was protected" and "The system prevented unauthorized access" garnered a mean of 4.37, showing uniform

agreement on the system's ability to protect user identities and manage access. In the meantime, the system's ability to provide content integrity was also positively rated, with a mean of 4.26, showing strong but slightly more diverse opinions on the system's ability to preserve the accuracy and reliability of stored academic content.

Table 19. Overall Result of System Evaluation

Indicators	Mean		Standard Deviation (SD)	
	Students	Instructors	Students	Instructors
Usability (Interaction Capability)	4.34	4.18	0.635	0. 934
Functional Suitability	4.39	4.37	0.616	0. 713
Reliability	4.36	4.29	0.582	0. 766
Performance Efficiency	4.54	4.39	0.545	0. 766
Perceived Ease of Use	4.36	4.38	0.597	0. 657
Security	4.41	4.38	0.521	0. 721
Overall	4. 40	4. 33	0.583	0.760

The overall evaluation revealed that both students and instructors strongly agreed with the quality and performance of the platform, with overall mean scores of 4.40 and 4.33, respectively. Among the specific areas, students gave the highest rating to Performance Efficiency with a mean of 4.54 and also instructor rated Performance Efficiency the highest, with a mean of 4.39, reflecting both their strong appreciation for the system’s responsiveness and processing speed. While both groups rated the system positively across all areas, students consistently gave slightly higher scores than instructors.

The standard deviation values supported these findings, showing the consistency of feedback within each group. Students demonstrated more

consistent responses, with an overall standard deviation average of 0.583, compared to 0.760 for instructors. The lowest standard deviation among students was observed in Security 0.521, suggesting that they strongly agreed on the platform's protective measures. In contrast, instructors showed the most consistent responses in Perceived Ease of Use, with a standard deviation of 0.657, reflecting shared experiences regarding user-friendliness. These results highlighted that although both groups found the system effective and reliable, students showed a slightly stronger and more uniform level of satisfaction across all measured aspects.

CHAPTER V

SUMMARY OF FINDINGS, CONCLUSIONS AND RECOMMENDATIONS

In this chapter, the researchers provided a summary of the study's findings, conclusions, and recommendations. Based on the findings, appropriate generalization was made, and conclusions and recommendations were presented accordingly.

Summary of Findings

The research assessed the Decision Support System (DSS) Archiving Hub, an AI-enabled academic repository created to computerize research management and facilitate archiving processes at North Eastern Mindanao State University (NEMSU) Tandag Campus. There were 363 participants in the evaluation of the system, including 344 students and 19 faculty members. Within the context of the ISO 25010 model, the evaluation targeted six main quality characteristics: Usability (Interaction Capability), Functional Suitability, Reliability, Performance Efficiency, Perceived Ease of Use, and Security. The findings showed high user satisfaction across all dimensions, with mean scores understood as "Strongly Agree."

Students assigned the system an overall mean rating of 4.40, and instructors a slightly lower but still positive overall mean of 4.33, both indicating a positive acceptance. Among all the dimensions, students rated Performance Efficiency highest at 4.54 due to quick response times and seamless document upload and retrieval processes. Instructors similarly rated Performance

Efficiency highly at 4.39, affirming the system's efficient management of data under different usage scenarios.

For Security, instructors and students provided almost similar ratings of 4.41 and 4.38, respectively, indicating mutual trust in the system's secure login mechanisms and data safety measures. Perceived Ease of Use also received excellent ratings, with 4.38 and 4.36, respectively, from instructors and students, recognising the easy-to-use interface, simple layout, and small learning curve of the system. Functional Suitability was also rated positively 4.39 by students and 4.37 by faculty in validating the system's success in providing expected functions, i.e., archiving, filtering, and searching thesis and feasibility studies.

Usability (Interaction Capability) received a mean of 4.34 from students and 4.18 from instructors, indicating the system facilitated easy user interaction and proper interaction, although instructor ratings were slightly more distributed (SD = 0.934), perhaps owing to varying usage patterns or technical experience. Reliability was also well appreciated, receiving 4.36 from students and 4.29 from instructors, indicating the system functioned across time with few errors or disruptions.

The statistics indicated slightly more uniform feedback from student users than instructors, as reflected by lower standard deviation values in most of the dimensions. This could imply the necessity of minor improvements specific to the instructor experience, like enhancing system feedback during multi-user use or adding administrative functionality. However, the results

confirmed that the DSS Archiving Hub achieved its specified objectives delivering an easy-to-use, responsive, and secure platform for managing academic research outputs. Such outcomes were consistent with prior work highlighting performance, usability, and security as key to effective academic information systems.

Conclusion

The study successfully responded to all the research questions enumerated in the Statement of the Problem through the development and testing of the DSS Archiving Hub. It discovered that both students and instructor before were severely difficulties by the lack of a systematic and centralized archiving system in storing, organizing, and retrieving theses and feasibility studies. These drawbacks tended to cause delays in academic processes, duplication of topics, and ineffective access to previous research outputs.

In order to address these issues, the study suggested and deployed the DSS Archiving Hub, a centralized digital platform meant to rigorously gather, maintain, and offer access to academic documents. The system allowed students to easily navigate and consult available research in the university, making literature review more effective and minimizing the potential for duplication of topics. Instructors were authorized to upload only accepted student research outputs upon defense, making sure that all archived documents were validated and quality-checked.

At the initial implementation of the system, the study was faced with significant limitations especially the requirement to manually enter metadata like the title, author, abstract, adviser, and other metadata fields in uploading PDF files. At that point, the system had only one AI feature aimed at research gap identification. It lacked automated systems to extract metadata and identify similarity, making it challenging for the users to locate similar studies or determine how many of the available works were similar to their selected topic.

These shortfalls were overcome through the incorporation of more AI-based features. An auto-fill system was integrated through natural language processing (NLP) to automatically extract important metadata from uploaded PDF documents, significantly reducing manual workload and input errors. Furthermore, a similarity detection feature was integrated using the K-Nearest Neighbors (KNN) algorithm. This allowed the system to analyze newly uploaded studies and identify content-related similarities with existing documents in the archive. It also generated statistical insights on how many related studies were present within the system, helping users and advisers make more informed decisions during research topic selection and reduce the likelihood of duplication.

By comprehensively addressing the issues outlined in the Statement of the Problem, the study not only identified the limitations of traditional archiving methods but also introduced a practical, efficient, and sustainable solution. The DSS Archiving Hub enhances the management of academic documents,

promotes research continuity, and supports a more organized and productive academic environment. Its successful implementation and positive reception position it as a valuable resource with potential for wider adoption across other academic units or institutions.

Recommendations

Based on the feedback and suggestions gathered from users of the Decision Support System (DSS) platform both students and instructors the following recommendations are proposed to enhance its functionality, usability, and overall effectiveness:

1. **Add a notification system** to alert instructors when they view documents.
2. **Integrate a plagiarism checker** to ensure the originality of submitted studies before archiving them.
3. **Expand export options** by allowing downloads not just in PDF but also in editable Word formats for easier citation and review.
4. **Develop a dedicated mobile application** to increase accessibility and convenience for both students and instructors when viewing or uploading research outputs.
5. **Optimize the system for low-bandwidth environments** and consider partial offline access or download options for users with limited internet connectivity.
6. **Allow bulk uploads** for easier processing of multiple approved works after defense week.

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